

One-dimensional simulation of geysering driven by entrapped-air release through water-filled vertical shafts using a decoupled two-fluid mixed-flow framework

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Abstract

Geysering in stormwater and combined-sewer systems occurs when entrapped air pockets are released through vertical shafts and expel the overlying water column above ground. System-scale prediction relies on reduced one-dimensional models, but most existing mixed-flow formulations prescribe the air phase through boundary data or lumped pocket laws, so the coupled air-pocket-water-column dynamics that select between benign venting and geysering are not part of the computed state. This paper applies a one-dimensional decoupled two-fluid cut-cell finite-volume framework, developed in a companion methods paper, to laboratory geysering experiments. The framework retains gas mass and gas momentum as conserved variables in stratified reaches, treats moving pressurized-stratified fronts as Regime-Transition Riemann problems, and represents the vertical shaft as the same one-dimensional two-fluid system rotated to the vertical, coupled to the horizontal conduit through a junction with local-loss closures. All simulations start from the experimental initial conditions and use one frozen solver

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configuration, without prescribed release rates, scripted geyser heights, or per-case fitting. The model is first evaluated against published ventilation-shaft experiments in which a pressurized air pocket is released toward a partially filled tower, with a wide-tower configuration in the non-geysering branch and a narrow-tower configuration in the geysering branch. The correct branch is selected in both configurations. In the non-geysering branch, the computed pressure-head plateau, terminal venting decay, air–water interface climb velocity, and maximum free-surface rise agree with the measurements. In the geysering branch, the pre-eruption stand of the tower column, the pressure plateau, the eruption to the shaft rim observed in every experimental repetition, the interface and free-surface climb velocities, and the sharp head collapse at pocket venting are reproduced, with the gas-phase event times uniformly early by about half a non-dimensional time unit. The model is then evaluated as a regime classifier against published horizontal-pipe/vertical-riser experiments, using the same frozen configuration under the fully coupled junction treatment: it reproduces both end members of the published two-parameter geysering criterion—the narrow-riser eruption with its pocket-compression pressure surge, and the benign venting of every wide-riser and small-pocket configuration—and the measured branch differentiation of the pipe-end pressure peak. Its classification errors over the experimental parameter sweep are strictly one-sided: near the regime boundary it under-predicts eruption, driving the water column high into the riser without expelling it, and it produces no false geyser. The conservative bias is traced to a single identified closure—the volume-neutral gas–water exchange at the junction mouth, whose demands in the two branches conflict—together

with one-dimensional limitations in junction transmission and annular film entrainment. A campaign against prototype-motivated junction-chamber experiments will complete the evaluation in a subsequent revision.

Keywords: geysering, entrapped air pocket, stormwater system, vertical shaft, two-fluid model, transient mixed flow

1. Introduction

Geysering denotes the violent expulsion of air–water mixtures through manholes, dropshafts, or ventilation shafts of stormwater and combined-sewer systems during rapid filling events. Field observations show mixtures rising tens of metres above ground, damaging manhole covers and endangering the public. Early explanations based on pure inertial oscillations of the water column were contradicted by field pressure records showing that the piezometric head near erupting shafts never approaches the ground surface [1]; the driving mechanism is instead the release of large entrapped air pockets through partially or fully water-filled vertical shafts [2, 1]. The laboratory experiments of Vasconcelos and Wright [2] established this air-release mechanism in a controlled single-shaft apparatus and remain the primary benchmark for geysering models. Subsequent experimental programmes examined the release of entrapped air from a horizontal pipe into a vertical shaft [3], water displacement and geysering in larger-scale shafts [4], stormwater geysers above junction chambers modelled on a prototype dropshaft [5], geysers through covered manholes [6], and mitigation measures for storm geysers [7]. The physical core of the event—a large gas pocket rising and expanding in a vertical water column until the expansion becomes violent—has also been

20 isolated in the multiphase-flow literature on Taylor bubble expansion [8].

21 Numerical prediction of geysering spans two extremes. Three-dimensional
22 volume-of-fluid simulations reproduce individual laboratory events in detail
23 [10], but their cost restricts them to single shafts and short time windows,
24 so they cannot be used for network-scale screening of hundreds of shafts
25 under many storm scenarios. At the other extreme, one-dimensional tran-
26 sient mixed-flow models are fast enough for system-scale simulation, but
27 most existing formulations treat the air phase incompletely. Preissmann-slot
28 and two-component-pressure formulations follow the pressurization of the liq-
29 uid phase [11], while entrapped air, when represented at all, enters through
30 lumped pocket laws, prescribed release rates, or boundary conditions. Rigid-
31 column two-phase models of the type used by Vasconcelos and Wright [2] to
32 interpret their own experiments prescribe the topology of the air pocket and
33 the water column in advance, which limits them to geometries where that
34 topology is known. Because the selection between benign venting and gey-
35 sering depends on the coupled dynamics of pocket compression, air intrusion
36 into the shaft, and counter-current air–water exchange at the shaft mouth,
37 a predictive one-dimensional model must carry the gas phase as part of the
38 computed state rather than as a prescribed forcing.

39 A companion methods paper [12] developed a one-dimensional conser-
40 vative finite-volume framework for transient closed-conduit mixed flows in
41 which the stratified reaches are described by a decoupled two-fluid system
42 that retains gas mass and gas momentum as conserved variables, liquid-full
43 reaches use an elastic-pipe water-hammer closure, and moving pressurized–
44 stratified fronts are treated as Regime-Transition Riemann problems embed-

45 ded in a conservative cut-cell front-tracking update. In that framework, air-
 46 pocket compression, gas transport, and interfacial-instability growth are out-
 47 comes of the resolved gas–liquid dynamics. The methods paper verified the
 48 framework on interfacial-wave, slug-formation, air-pocket pressure, and in-
 49 tegrated mixed-flow benchmarks, and included a schematic one-dimensional
 50 geysering-onset test; it did not, however, attempt a quantitative comparison
 51 against geysering experiments.

52 The present paper provides that application-level assessment. The frame-
 53 work of [12] is applied, unchanged in its formulation, to laboratory geysering
 54 experiments in which a vertical shaft is coupled to a horizontal conduit and
 55 an entrapped air pocket is released toward the shaft. The vertical shaft is dis-
 56 cretized as the same one-dimensional two-fluid system rotated to the vertical
 57 and coupled to the horizontal pipe through a junction closure with physically
 58 motivated local losses. All regime selection—whether the free surface reaches
 59 the shaft rim before the rising air–water interface catches it—emerges from
 60 the computed state.

61 The evaluation is organized as test campaigns against published exper-
 62 iments, each contributing a different kind of evidence. The first campaign
 63 is the water-filled ventilation-shaft experiment of Vasconcelos and Wright
 64 [2], from which two configurations are selected to span the two observed
 65 regimes: a large-tower case in the non-geysering branch and a small-tower
 66 case in the geysering branch, testing the time-resolved pressure and interface
 67 histories. The second campaign is the horizontal-pipe/vertical-riser exper-
 68 iment of Cong, Chan and Lee [3], whose systematic parameter study con-
 69 densed the geysering outcome into an explicit two-parameter criterion; it

70 tests the model as a regime classifier across the high-speed-camera series and
71 a 63-configuration sweep of the experimental parameter ranges. All computa-
72 tions use a frozen solver configuration and start from the experimental initial
73 conditions, without prescribed release rates, scripted geyser heights, or per-
74 case parameter fitting. A third campaign against the prototype-motivated
75 junction-chamber experiments of Liu, Shao and Zhu [5] tests branch selection
76 with a downstream tail boundary, eruption height against peak pressure, and
77 the transient mass-oscillation model of their Series C cases; other experimen-
78 tal studies [4, 6, 7, 9] serve as corroborating references.

79 The remainder of the paper is organized as follows. Section 2 summarizes
80 the decoupled two-fluid framework and describes the vertical-shaft branch
81 and junction closures used for geysering configurations. Section 3 presents
82 the test cases and the comparison with the experiments. Section 4 discusses
83 the capabilities and limitations identified by the comparison, and Section 5
84 concludes.

85 2. Model framework

86 This section summarizes the one-dimensional framework of the companion
87 methods paper [12] to the extent needed to interpret the geysering compu-
88 tations, and then describes the vertical-shaft branch, junction closures, and
89 trapped-pocket treatment used in the geysering configurations. The deriva-
90 tions, the Regime-Transition Riemann solver, and the conservative cut-cell
91 front-tracking algorithm are given in [12] and are not repeated here.

92 *2.1. Decoupled two-fluid mixed-flow formulation*

93 The framework describes a closed conduit as a union of liquid-full (pres-
 94 surized) and stratified gas-liquid reaches separated by moving regime fronts.
 95 In stratified reaches the flow is governed by the decoupled two-fluid balance
 96 law of [12],

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = \mathbf{S}, \quad (1)$$

97 with conservative state, flux, and regular source vectors

$$\mathbf{U} = \begin{bmatrix} A_g \rho_g \\ \rho_g Q_g \\ A_l \\ Q_l \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \rho_g Q_g \\ \rho_g Q_g^2 / A_g + P_g A_g \\ Q_l \\ Q_l^2 / A_l + \frac{1}{2} \Lambda_d A_l^2 \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} 0 \\ P_g \frac{\partial A_g}{\partial x} - A_g \rho_g g \cos \theta \frac{\partial h_l}{\partial x} + \mathcal{S}_g \\ 0 \\ \mathcal{S}_d \end{bmatrix}, \quad (2)$$

98 where A_g and A_l are the gas and liquid cross-sectional areas, ρ_g the gas den-
 99 sity, Q_g and Q_l the gas and liquid volumetric flow rates, P_g the gas pressure,
 100 h_l the liquid depth, θ the pipe inclination, and $\mathcal{S}_g$, $\mathcal{S}_d$ the regular gas- and
 101 liquid-momentum sources containing axial gravity, wall friction, and inter-
 102 facial shear; the shear closures follow standard stratified-flow correlations
 103 [13, 14]. Gas mass and gas momentum are conserved variables, so gas trans-
 104 port, gas inertia, gas compressibility, and shear-driven interfacial growth are
 105 part of the computed state. The restoring coefficient in the liquid momentum
 106 flux is

$$\Lambda_d = \underbrace{\frac{2gH_g}{A_l}}_{\text{gas pressure}} + \underbrace{\frac{\rho_l - \rho_g}{\rho_l} g \zeta}_{\text{hydrostatic-buoyancy}} - \underbrace{\frac{\rho_g(u_g - u_l)^2}{\rho_l A_g}}_{\text{IKH slip}}, \quad (3)$$

107 which combines the gas-pressure, hydrostatic-buoyancy, and inviscid Kelvin-
 108 Helmholtz (IKH) slip contributions; here H_g is the gas pressure head, u_g

and u_l the phase velocities, and $\zeta = \cos \theta / [D \sin(\gamma/2)]$ a circular-section geometric factor with D the pipe diameter and γ the liquid central angle. The eigenvalues of the decoupled liquid flux Jacobian,

$$\lambda^\pm = u_l \pm \sqrt{\Lambda_d A_l}, \quad (4)$$

show that the liquid subsystem is strictly hyperbolic for $\Lambda_d > 0$ and reaches the neutral interfacial-instability (IKH) threshold at $\Lambda_d = 0$; beyond it, interfacial waves grow and liquid bridging can occur.

In liquid-full reaches the gas phase is absent and the pressurized branch is governed by the two-variable system

$$\frac{\partial A_l}{\partial t} + \frac{\partial Q_l}{\partial x} = 0, \quad \frac{\partial Q_l}{\partial t} + \frac{\partial}{\partial x} \left(\frac{Q_l^2}{A_l} + g A_f h_s \right) = \mathcal{S}_p, \quad (5)$$

closed by the elastic-pipe relation

$$P - P_{\text{ref}} = \rho_l a^2 \frac{A_l - A_f}{A_f}, \quad h_s = \frac{P - P_{\text{ref}}}{\rho_l g}, \quad (6)$$

with a the (numerical) pressure-wave speed, A_f the undeformed pipe area, h_s the piezometric head, and $\mathcal{S}_p$ the regular pressurized-branch source; this branch recovers the water-hammer limit. In limiting cases the same area-variable form of Eqs. (1)–(6) recovers water hammer, open-channel gravity flow (with $\Lambda_d \rightarrow g \cos \theta / B_l$, where B_l is the free-surface top width), interfacial-instability dynamics, and the near-full mixed-flow balance [12].

Moving pressurized–stratified fronts are internal boundaries between the two-variable system (5) and the four-variable system (1). Their local closure and the conservative front-tracking update are summarized in Sections 2.2–2.3, so that the computations reported here can be interpreted and reproduced from the present paper alone; the derivations, the construction of the

129 active-set solver, and the verification of the scheme on canonical benchmarks
 130 are given in [12].

131 Two properties of this formulation matter for geysering. First, because gas
 132 mass is conserved per computational region, entrapped pockets are identified
 133 as connected gas regions bounded by liquid seals, and their pressure follows
 134 from an isothermal equation of state $P_g = m_g RT/V_g$ evaluated with the
 135 current pocket mass m_g and volume V_g ; no polytropic pocket law with a
 136 prescribed volume history is imposed. Second, because the stratified branch
 137 carries gas momentum, air intrusion into a shaft and counter-current air-
 138 water exchange at the shaft mouth are resolved rather than prescribed.

139 2.2. Regime-Transition Riemann closure at moving fronts

140 At a moving pressurized-stratified front $x = x_\Gamma(t)$ with speed $w_\Gamma = \dot{x}_\Gamma$,
 141 the local coordinate is oriented so that the pressurized branch lies on the left
 142 and the stratified branch on the right; the opposite case follows by reflection.
 143 Only liquid mass and liquid momentum are common to both branches, and
 144 the ideal front has zero thickness and stores neither. Integrating the mixed-
 145 regime liquid balance over a control volume shrinking onto the front therefore
 146 gives the moving Rankine-Hugoniot conditions

$$\begin{aligned} Q_{p,\Gamma} - Q_{f,\Gamma} &= w_\Gamma (A_{p,\Gamma} - A_{l,\Gamma}), \\ \left(\frac{Q_{p,\Gamma}^2}{A_{p,\Gamma}} + g A_f h_{s,\Gamma} \right) - \left(\frac{Q_{f,\Gamma}^2}{A_{l,\Gamma}} + \frac{1}{2} \Lambda_{d,\Gamma} A_{l,\Gamma}^2 \right) &= w_\Gamma (Q_{p,\Gamma} - Q_{f,\Gamma}), \end{aligned} \quad (7)$$

147 where the subscripts p and f denote the pressurized trace and the liquid
 148 projection of the stratified trace at the front, and $h_{s,\Gamma}$ follows from the elastic
 149 closure (6). The pressurized side is liquid-full, so gas does not cross the front;

150 the moving-frame gas condition gives

$$u_{g,\Gamma} = w_\Gamma. \quad (8)$$

151 Because the water-hammer speed a is much larger than mixed-flow front
 152 speeds, one pressurized acoustic characteristic always reaches the front, sup-
 153 plying the compatibility relation

$$u_{p,\Gamma} - u_{p,L} + \frac{g}{a} (h_{s,\Gamma} - h_{s,L}) = 0, \quad (9)$$

154 with L the adjacent pressurized state. Equations (7)–(9) do not determine all
 155 interface unknowns; the remaining information depends on the position of w_Γ
 156 relative to the stratified liquid signal speeds $s_\pm = u_{l,R} \pm c_{l,R}$, $c_{l,R} = \sqrt{\Lambda_{d,R} A_R}$,
 157 which partition the admissible front speeds into the slow ($w < s_-$), middle
 158 ($s_- \leq w \leq s_+$), and fast ($w > s_+$) active sets.

159 Because the correct active set is not known before w_Γ is computed, and
 160 branch-dependent nonlinear iteration at every front is fragile in transient
 161 computations, the closure of [12] is a fixed-cost one-step construction. Substi-
 162 tuting the compatibility relation (9) and the mass jump into the momentum
 163 jump reduces the closure, for any candidate speed w , to a scalar quadratic
 164 for $A_{p,\Gamma}(w)$ and a scalar momentum residual $R(w)$. A finite-area predictor
 165 built from the adjacent states,

$$w_0 = \frac{A_{p,L} u_{p,L} - A_R u_{l,R}}{A_{p,L} - A_R}, \quad (10)$$

166 is projected onto each (slightly shrunk) active-set interval, one closed-form
 167 Newton-type residual correction is applied per interval, and the accepted
 168 candidate is the one minimizing a penalized criterion combining the Rankine–
 169 Hugoniot residual, the distance to its active set, and positivity penalties. No

170 Bernoulli-type energy relation is imposed at the front. The selected trace
 171 defines the moving-frame flux pair $\mathbf{G}_{p,\Gamma} = \mathbf{F}_{p,\Gamma} - w_\Gamma \mathbf{U}_{p,\Gamma}$ and $\mathbf{G}_{f,\Gamma} = \mathbf{F}_{f,\Gamma} -$
 172 $w_\Gamma \mathbf{U}_{f,\Gamma}$ passed to the cut-cell update below.

173 *2.3. Finite-volume update and cut-cell front tracking*

174 Away from fronts, both branches are advanced by the same explicit finite-
 175 volume machinery: MUSCL–TVD primitive reconstruction with the minmod
 176 limiter, branch-dependent Rusanov fluxes (scalar dissipation $\max(|u_p| + a)$ on
 177 the pressurized branch; block-diagonal dissipation with gas speed $|u_g| + c_g$ and
 178 stratified liquid speed $|u_l| + c_\Lambda$, $c_\Lambda = \sqrt{\max(\Lambda_d A_l, 0) + c_{\text{num}}^2}$, on the stratified
 179 branch), unsplit cell-centered sources, and SSP–RK2 time integration. The
 180 time step is the minimum of the branch CFL limits and a front-motion restric-
 181 tion $\Delta t_\Gamma = \frac{1}{2} \min(\Delta x_p, \Delta x_f) / |w_\Gamma^n|$, which prevents the front from sweeping
 182 more than half a cell per step.

183 The cell hosting a front is divided into pressurized and stratified sub-
 184 cells of lengths ℓ_p and ℓ_f with $\ell_p + \ell_f = \Delta x$. At the beginning of each
 185 step the adjacent branch states are passed to the closure of Section 2.2; the
 186 front is then advanced as $x_\Gamma^{n+1} = x_\Gamma^n + w_\Gamma \Delta t$, and each sub-cell is updated
 187 by its own branch flux at the outer face and the moving-frame interface
 188 flux at the front. Because both sub-cell balances use the same w_Γ and the
 189 same interface trace, the interface fluxes and swept-volume contributions
 190 cancel exactly for the conserved liquid components and gas mass: the front
 191 neither creates nor destroys liquid or gas. When the front crosses a cell
 192 face, the affected cells are remapped conservatively (event-based remap),
 193 and a bounded, localized residual relaxation corrects the remaining interface
 194 mismatch after an accepted step. Trapped gas regions are identified after each

195 step as connected gas volumes bounded by liquid seals, and their pressures
 196 are evaluated from the isothermal equation of state on the current pocket
 197 mass and volume.

198 *2.4. Vertical-shaft branch*

199 A vertical shaft (ventilation tower, riser, or dropshaft barrel) is discretized
 200 as a separate one-dimensional finite-volume branch using the same area–
 201 momentum variables as the horizontal conduit, with the pipe inclination
 202 set to $\theta = 90^\circ$ so that gravity acts axially. The shaft top is open to the
 203 atmosphere; the free surface inside the shaft and any gas core penetrating
 204 below it are represented by the axial profile of the gas volume fraction $\alpha_g(z)$,
 205 not by a prescribed bubble shape. The shaft state therefore distinguishes
 206 the free-surface elevation Y_{fs} (top of the continuous liquid column) from
 207 the air–water interface elevation Y_{int} (nose of the gas core rising through the
 208 column), which are exactly the two trajectories reported in shaft experiments.
 209 Geysering is diagnosed, following the experimental definition of [2], when Y_{fs}
 210 reaches the shaft top before Y_{int} catches the free surface.

211 *2.5. Junction and local-loss closures*

212 The shaft base is coupled to the horizontal conduit through a junction
 213 closure driven by the local pipe state and the trapped-pocket pressure. The
 214 junction exchange uses the pocket equation-of-state static head as the driving
 215 head, with the following local-loss elements retained from the frozen config-
 216 uration used for all runs in Section 3:

- 217 • a junction entry/exit loss coefficient $K_j = 0.75$ on the through-flow
 218 between pipe and shaft base;

- 219 • an additional churn loss $K_g = 8.0$ applied while the junction carries
 220 gas, representing the strongly dissipative counter-current air-up/water-
 221 down exchange (“glugging”) at the shaft mouth;
- 222 • a butterfly-valve loss $K_v = 2.0$ at the release valve, whose disc remains
 223 in the flow after opening, with a finite hand-opening time $t_v = 0.25$ s
 224 represented by throttling the loss coefficient during opening;
- 225 • a counter-current flooding (CCFL) limit of Wallis type on reverse liquid
 226 drainage at the shaft top while gas is venting, with limiting velocity
 227 scale $0.5\sqrt{gD_t}$, where D_t is the shaft diameter.

228 Three further elements govern the shaft base in the driven regime, i.e.
 229 when the trapped pocket holds an overpressure margin over the shaft column
 230 ($H_{a0} > Y_{fs,0}$) and gas has reached the junction; they are stated here and
 231 motivated with the narrow-tower test in Section 3.1.2: (i) the shaft base is
 232 driven by the pipe *crown* pressure (the elevation at which the shaft physically
 233 taps the pipe), one bore below the invert piezometric line while the junction
 234 is water-sealed; (ii) the shaft gas core connected to the pocket is pocket-fed—
 235 mouth influx raises the core nose, by the same fill-to-body-level closure used
 236 for the crown cavity in the horizontal pipe—rather than accumulating at the
 237 base cell under dispersed-bubble drag; and (iii) the entry gas flux is bounded
 238 by the Wallis counter-current flooding scale $j_g \leq C_w^2 \sqrt{gD_t \Delta\rho/\rho_l}$ with $C_w =$
 239 0.9. These closures use geometric facts and literature-scale constants; in
 240 the undriven regime (Test A of Campaign 1) they are inactive and the base
 241 retains the invert-datum coupling, a dual treatment whose consolidation is
 242 discussed in Section 4.

243 Once a trapped pocket becomes connected to the shaft and gas breaks
 244 through the liquid seal, venting latches on: the pocket continues to dis-
 245 charge through the shaft and is not re-sealed by intermittent slugs. This
 246 latch reflects the experimental observation that after breakthrough the re-
 247 lease proceeds as a sustained, if oscillatory, discharge rather than a sequence
 248 of independent seal events.

249 *2.6. Numerical configuration*

250 All computations in Section 3 use one frozen solver configuration: horizon-
 251 tal grid spacing $\Delta s = 0.02$ m, shaft grid spacing $\Delta z = 0.01$ m, CFL number
 252 0.35, and a reduced numerical elastic wave speed $a = 28$ m/s in the liquid-
 253 full branch. The reduced wave speed keeps the liquid-full reaches effectively
 254 incompressible on the pocket and shaft timescales (the gravity-wave scale is
 255 $\sqrt{gD} \approx 1$ m/s) while avoiding stiff acoustic ringing at grid-scale micro-voids;
 256 the geysering dynamics studied here are pocket-compression and gravity-
 257 wave driven, not water-hammer driven. Gas thermodynamics is isothermal
 258 with $R = 287$ J/(kg K) and $T = 293$ K, and the liquid density is $\rho_l =$
 259 998 kg/m³. No loss coefficient or numerical parameter is adjusted between
 260 test cases; the only case-dependent element is the regime-conditioned shaft-
 261 base treatment of Section 2.5 (crown-datum, pocket-fed, flooding-limited in
 262 the driven regime; invert-datum otherwise), which switches on the physical
 263 state, not on the test identity.

264 **3. Test cases**

265 The evaluation is organized as a sequence of test campaigns against pub-
 266 lished geysering experiments. Within each campaign, individual configu-

267 rations are selected to span the experimentally observed regimes, and all
 268 configurations are computed with the single frozen solver configuration of
 269 Section 2.6. Simulations start from the documented experimental initial
 270 conditions; the shaft motion, the air release, and the regime selection are
 271 outcomes of the computation. No prescribed release rate, scripted geyser
 272 height, overflow accumulator, or per-case parameter fitting is used.

273 Three campaigns are selected from the experimental geysering literature
 274 so that each contributes a different kind of evidence. Campaign 1 (Sec-
 275 tion 3.1) reproduces the water-filled ventilation-shaft experiments of Vas-
 276 concelos and Wright [2]—the most widely cited controlled geysering exper-
 277 iments and the standard benchmark for geysering models—and tests the
 278 time-resolved histories in both regimes: Test A in the non-geysering branch
 279 and Test B in the geysering branch. Campaign 2 (Section 3.2) reproduces
 280 the horizontal-pipe/vertical-riser experiments of Cong, Chan and Lee [3], the
 281 only campaign that condenses its outcome into an explicit two-parameter
 282 geysering criterion; it therefore tests the model as a regime classifier across a
 283 systematic parameter sweep rather than in individual histories. Campaign 3
 284 (Section 3.3) addresses the prototype-motivated junction-chamber experi-
 285 ments of Liu, Shao and Zhu [5], which connect the laboratory benchmarks to
 286 a field-scale dropshaft geometry. Other experimental studies—the field ob-
 287 servations of [1], the larger-scale releases of [4] designed for three-dimensional
 288 CFD calibration, the covered-manhole geysers of [6], the mitigation study of
 289 [7], and the vertical-pipe violent geysers of [9]—are used as corroborating ref-
 290 erences rather than as reproduction targets, either because their geometry is
 291 not controlled at the level needed for one-dimensional comparison or because

292 their focus (cover dynamics, mitigation devices, flashing-driven eruption) lies
 293 outside the entrapped-air release mechanism studied here.

294 *3.1. Campaign 1: water-filled ventilation shafts (Vasconcelos and Wright,*
 295 *2011)*

296 The apparatus of [2] consists of a horizontal acrylic pipe of diameter $D =$
 297 0.094 m with both far ends closed, a ventilation tower of length $L = 0.610$ m
 298 and variable diameter D_t mounted on a coupling 3.516 m from the upstream
 299 end, and an upstream air chamber of length 0.546 m separated from the
 300 water-filled reach by a butterfly valve (Fig. 1). The air chamber is pressurized
 301 to an initial gauge head H_{a0} , the tower is partially filled to an initial level
 302 $Y_{fs,0}$, and the run starts when the valve is opened by hand in less than
 303 one second. The pressurized pocket then propagates along the pipe crown
 304 toward the tower and vents through the water column inside it. A pressure
 305 transducer is located 1.07 m downstream of the valve. The experimental
 306 programme varied $D_t \in \{57.1, 44.4, 25.4, 12.7\}$ mm ($D_t/D = 0.607$ – 0.135),
 307 $H_{a0} \in \{0.305, 0.610, 0.915\}$ m, and $Y_{fs,0} \in \{0.254, 0.356, 0.457\}$ m. The
 308 controlling variable was found to be the tower diameter: for $D_t/D = 0.607$
 309 the free-surface rise never exceeded 10% of L , whereas for $D_t/D = 0.135$ the
 310 free surface reached the tower top in every repetition.

311 Two configurations are selected to span the two regimes (Table 1). Fol-
 312 lowing [2], results are non-dimensionalized by the tower length and the tower
 313 gravity-velocity scale: elevations and pressure heads as $Y^* = Y/L$ and
 314 $H^* = H/L$, and time as $T^* = t\sqrt{gD_t}/L$. Geysering is defined as the free sur-
 315 face Y_{fs}^* reaching the tower top ($Y^* = 1$) before the rising air–water interface
 316 Y_{int}^* catches the free surface.

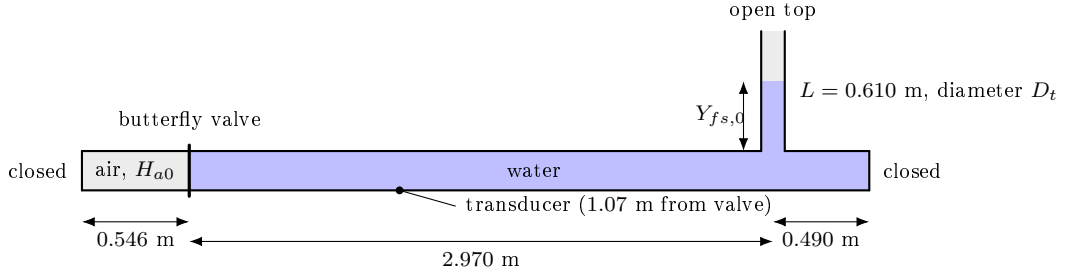


Figure 1: Schematic of the Vasconcelos and Wright [2] apparatus as discretized in the model (not to scale in the transverse direction). The horizontal pipe ($D = 0.094$ m) is closed at both far ends; the upstream chamber holds the pressurized air pocket; the ventilation tower (length $L = 0.610$ m, diameter D_t) is open at the top and partially filled to $Y_{fs,0}$.

Table 1: Selected test configurations from the experimental matrix of [2].

Test	D_t (mm)	D_t/D	H_{a0} (m)	$Y_{fs,0}$ (m)	$L/\sqrt{gD_t}$ (s)	Observed regime
A	57.1	0.607	0.305	0.356	0.815	no geysering
B	12.7	0.135	0.610	0.356	1.728	geysering (every run)

317 The model geometry follows the apparatus exactly: upstream air chamber
 318 0.546 m (the initial pocket volume, $3.79 \times 10^{-3} \text{ m}^3$), middle pipe 2.970 m,
 319 downstream closed pipe 0.490 m, tower at $x = 3.516 \text{ m}$, and the transducer
 320 sampled at $x = 1.616 \text{ m}$. The initial pocket pressure is $P_{a0} = P_{\text{atm}} + \rho_l g H_{a0}$;
 321 the pipe downstream of the valve and the tower below $Y_{fs,0}$ are initially at
 322 rest and water-filled. The experimental curves used for comparison were
 323 digitized from the published figures of [2] (pressure head from Figs. 5 and 6;
 324 free-surface and interface trajectories from Figs. 7 and 8, three experimental
 325 repetitions each); the digitized repetitions are shown either individually or
 326 as a band.

327 3.1.1. Test A: large tower, non-geysering branch ($D_t/D = 0.607$)

328 In the experiment this configuration vents the pocket without geysering:
 329 the tower free surface rises by less than 10% of L , the transducer head holds a
 330 long plateau near $H^* \approx 0.54$ while the pocket discharges, and the head then
 331 decays to zero once the pocket is exhausted. The computed run reproduces
 332 this branch. The free surface peaks at $Y_{fs}^* = 0.66$ (measured maximum 0.63),
 333 the rising gas core catches the free surface inside the tower, and no overflow
 334 occurs.

335 Figure 2 compares the transducer pressure head. The model reproduces
 336 the plateau level (cycle-averaged $H^* = 0.51$ against the digitized experimen-
 337 tal plateau $H^* = 0.54$) over the full venting window $1 \lesssim T^* \lesssim 7.5$, and
 338 reproduces the terminal decay associated with pocket exhaustion, beginning
 339 at $T^* \approx 8.9$ in the model against 8.3 in the experiment. Two differences are
 340 visible. First, in $0 < T^* \lesssim 5$ the raw model trace carries a water-column-
 341 pocket spring oscillation with a period of about 0.7 s that decays by $t \approx 4$ s;

342 mass-conservation and phase checks identify it as a genuine volume–pressure
 343 oscillation of the release piston rather than a numerical artefact, but the
 344 experimental traces are more strongly damped. The plotted model curve is
 345 therefore cycle-averaged over a 0.8 s window. Second, the venting stage runs
 346 late and more gradually in the model: the measured head starts falling at
 347 $T^* \approx 8.3$ and reaches zero near 9.3, while the model holds its plateau to
 348 $T^* \approx 8.9$ and decays to about $H^* \approx 0.21$ by the end of the window. Both
 349 features reflect the slower discharge of the pocket through the tower once the
 350 (dissipative) cavity celerity feeds the vent at the reduced rate; the arrival-
 351 stage timing gains of the celerity value are traded against the venting-stage
 352 lag (Section 4).

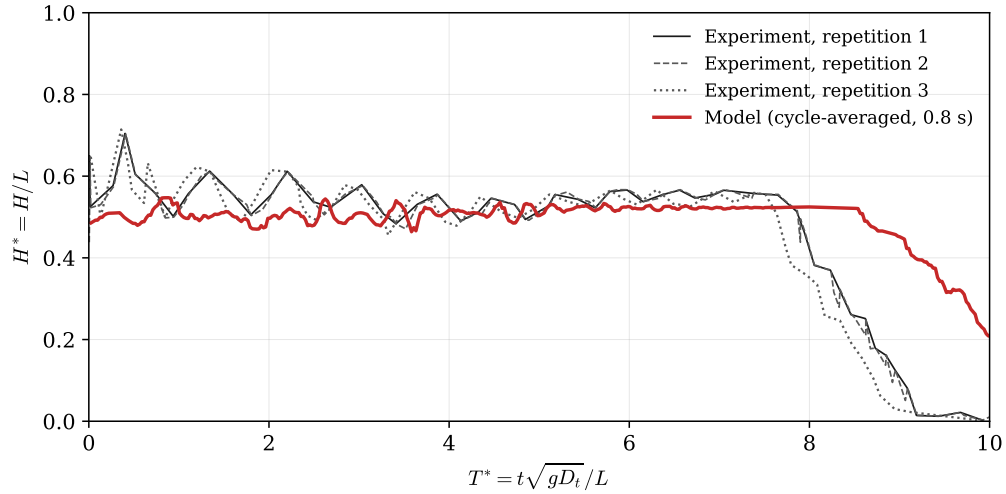


Figure 2: Test A ($D_t/D = 0.607$, $H_{a0} = 0.305$ m, $Y_{fs,0} = 0.356$ m): transducer pressure head H^* against T^* . Black lines: the three experimental repetitions of [2], digitized point by point from their Fig. 5; red line: model, cycle-averaged over a 0.8 s window.

353 Figure 3 compares the tower free-surface and air–water-interface trajec-

354 tories in the published observation window ($T^* = 7\text{--}10$, Fig. 7 of [2]). The
 355 model holds the free surface near $Y_{fs}^* \approx 0.6$ throughout the venting plateau,
 356 consistent with the measured level, and produces the interface climb through
 357 the tower with the correct slope: the fitted non-dimensional interface veloc-
 358 ity is $V_{int}^* = 0.41$, against the experimental Table-2 average of 0.39 and a
 359 photograph-based estimate of 0.50 for this tower size. The interface leaves
 360 the tower base at $T^* = 7.5$, inside the spread of the three experimental rep-
 361 etitions (climb onsets at 7.3, 7.8, and 7.9), and catches the free surface at
 362 $T^* = 8.8$ against the measured 8.4 (Table 2). Two systematic differences in
 363 the level histories are noted. First, the measured free surface creeps upward
 364 during the interface climb (from 0.59 to 0.65), which [2] attribute to the ex-
 365 pansion of the depressurizing pocket as it occupies the tower; the computed
 366 level stays at its plateau value because the volume-neutral counter-current
 367 exchange at the mouth transits gas through the standing column without
 368 net volume addition, so the computed climb-phase free-surface velocity is
 369 $V_{fs}^* \approx 0$ against the measured 0.048 (the measured maximum level itself is
 370 matched: $Y_{fs,\max}^* = 0.66$ computed against 0.63 measured). Second, after the
 371 interface reaches the free surface, the measured tower level stays near $0.6L$
 372 to the end of the observation window, while the model level decays faster
 373 once venting is established; the counter-current flooding limit at the tower
 374 mouth (Section 2.5) bounds but does not eliminate this too-fast drainage.
 375 Both differences trace to the same film-flow pocket topology discussed in
 376 Section 4.

377 Figure 4 places the published photographic sequence and the computed
 378 tower snapshots together at the same 0.14 s spacing during the interface

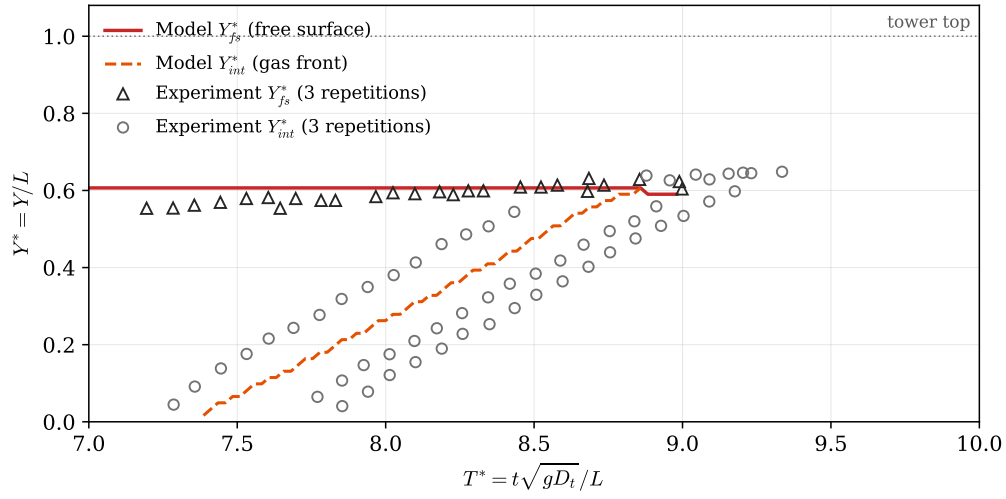


Figure 3: Test A: tower free-surface elevation Y_{fs}^* (solid) and air–water interface elevation Y_{int}^* (dashed) against T^* in the published observation window ($T^* = 7\text{--}10$). Symbols: the three experimental repetitions of [2], digitized from their Fig. 7 (triangles: free surface; circles: interface). The model gas-front curve ends where it catches the free surface (beyond that instant the two interfaces merge and a distinct front no longer exists); the model free-surface curve is shown over the extent of the experimental coverage.

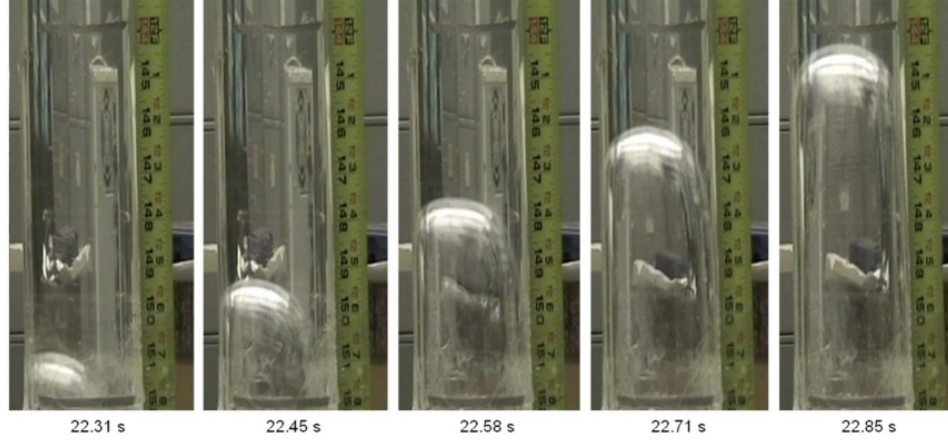
379 climb. The gas core width in the computed row is drawn from the local two-
380 fluid gas fraction without thresholding. The computed front positions track
381 the photographed front to within 0.03 m over the sequence, and the mean
382 front speed over the photographed interval is 0.375 m/s in the experiment-
383 based estimate against 0.31 m/s in the model fit window.

Table 2: Test A: measured against computed quantities. Times are non-dimensional ($T^* = t\sqrt{gD_t}/L$); velocities are normalized by $\sqrt{gD_t}$. Experimental values are digitized from [2] (Figs. 4, 5, 7 and Table 2).

Quantity	Experiment	Model
Regime	no geysering	no geysering
Maximum free-surface rise $Y_{fs,max}^*$	0.63	0.66
Pressure-head plateau H^*	0.54	0.51
Onset of terminal head decay T^*	8.3	8.9
Interface liftoff from tower base T^*	7.3/7.8/7.9 (3 reps)	7.5
Interface catches free surface T^*	8.4	8.8
Interface climb velocity V_{int}^*	0.39 (Table 2), 0.50 (Fig. 4)	0.41
Climb-phase free-surface velocity V_{fs}^*	0.048	≈ 0

384 Finally, Fig. 5 compares the present model directly against the numerical
385 model of [2] themselves (their two-phase-analysis (TPA) model, a lumped
386 pocket formulation with a Batchelor-type laminar film-flow closure). That
387 comparison is reported for the neighbouring configuration $D_t/D = 0.607$,
388 $H_{a0} = 0.305$ m, $Y_{fs,0} = 0.254$ m—the same tower and pocket pressure as
389 Test A with a lower initial level—so the frozen solver is re-run at that initial
390 condition. The TPA-model curves and the experimental traces were digitized

(a) Experiment: Vasconcelos and Wright (2011), Fig. 4



(b) Present two-fluid model at the same 0.14 s spacing

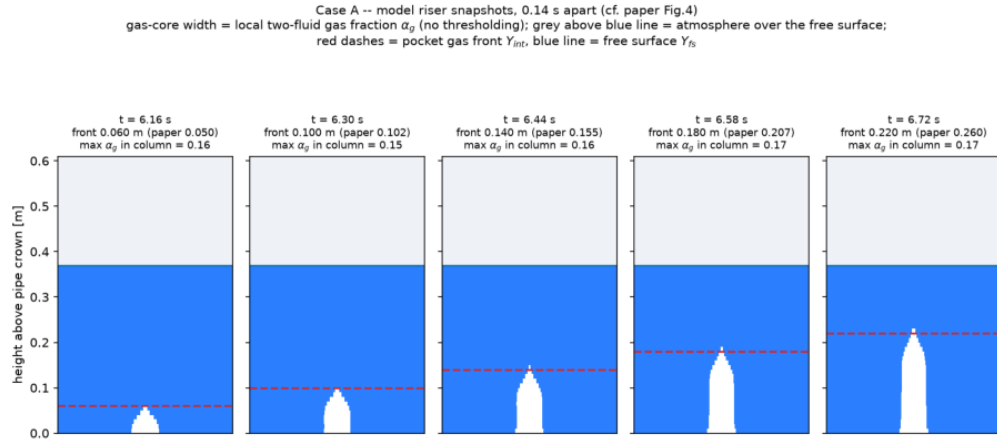


Figure 4: Test A, direct comparison with Fig. 4 of [2]. Top: the five published experimental photographs. Bottom: computed tower fields at the same 0.14 s spacing. Blue: water; white: gas core (local two-fluid gas fraction); grey: atmosphere above the free surface. The red dashed line marks the computed pocket gas front Y_{int} and the panel captions give the computed and photographed front elevations.

391 from Fig. 10 of [2] and are replotted here together with the present compu-
 392 tation for the two panels that carry the discussion: the interface trajectory
 393 and the transducer head. Because the valve is opened by hand, a rigid time
 394 shift between the model and the published time axis would be legitimate;
 395 none is needed for the interface panel (the computed climb onset sits +0.2
 396 from the published model curve, inside the repetition scatter) and none is
 397 applied. The interface trajectory Y_{int}^* reproduces the climb slope and span of
 398 both the measurements and the TPA model; the interface-velocity plateau
 399 ($V_{int}^* \approx 0.37$ computed, ≈ 0.4 TPA and measured, from the velocity panels
 400 of their Fig. 10) agrees; the pressure-head decay runs about 0.7 time units
 401 late (the venting-stage lag of the reduced cavity feed, as in Test A); and the
 402 head panel shows the structural difference already noted: the measured head
 403 and the TPA model decay *during* the climb, following the shrinking water
 404 column above the pocket nose, while the present model holds its plateau until
 405 breakthrough and then collapses. The TPA model obtains the during-climb
 406 decay by construction—its pocket is a lumped volume whose pressure is tied
 407 to the column above the nose through the film-flow closure—whereas in the
 408 present field formulation the tower base carries the weight of the full standing
 409 column as long as the gas transits it as a dispersed phase. Section 4 discusses
 410 this topological limitation and the closure experiment performed on it.

411 For completeness, Fig. 6 retains all five panels of the published Fig. 10
 412 comparison rather than only the two panels replotted above. The complete
 413 view makes two additional findings explicit. The computed interface-velocity
 414 level is comparable with the measurements and TPA result during the co-
 415 herent climb, whereas the computed free-surface velocity lacks the measured

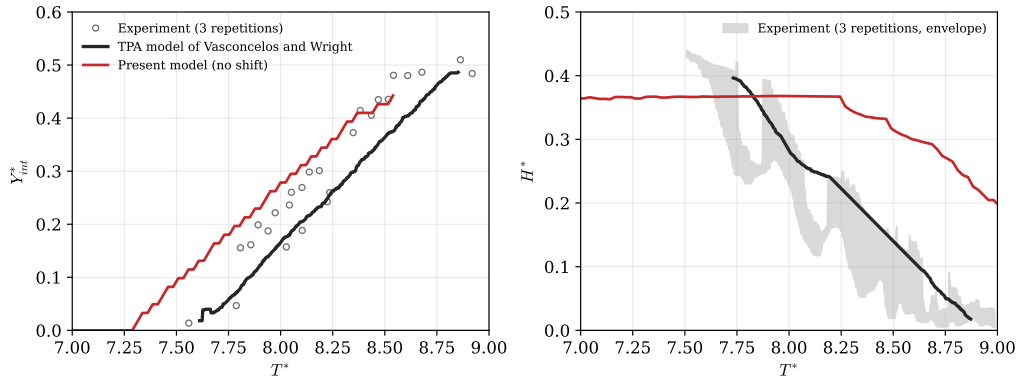


Figure 5: Campaign 1, configuration $D_t/D = 0.607$, $H_{a0} = 0.305$ m, $Y_{fs,0} = 0.254$ m: present model (red) against the TPA model of [2] (black) and their measurements (grey), both digitized from Fig. 10 of [2]. Left: air–water interface elevation Y_{int}^* (circles: experimental repetitions). Right: transducer pressure head H^* (shaded: envelope of the three experimental repetitions). No time shift is applied to the present-model histories (see text).

oscillatory positive pulses. The latter is the velocity-space counterpart of the
volume-neutral mouth-exchange limitation already visible in the free-surface
trajectory.

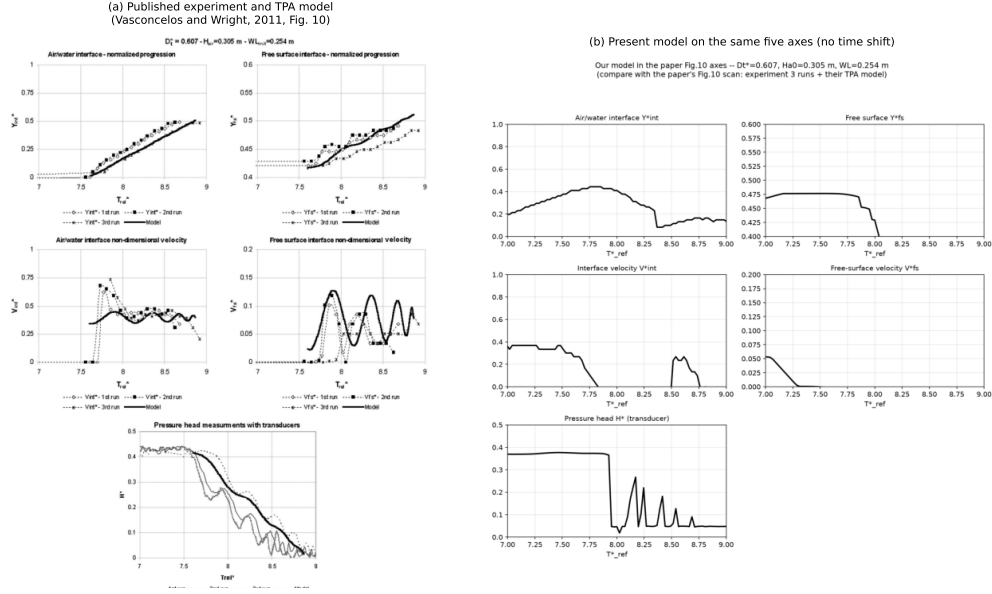


Figure 6: Campaign 1, neighbouring $Y_{fs,0} = 0.254$ m configuration: complete five-panel comparison corresponding to Fig. 10 of [2]. Left: published measurements and TPA model. Right: present model on the same axes, with no time shift. The panels show Y_{int}^* , Y_{fs}^* , V_{int}^* , V_{fs}^* and H^* , respectively. The reproduced source panel is included here for direct visual verification; the two quantitatively discussed traces are independently digitized and redrawn in Fig. 5.

3.1.2. Test B: small tower, geysering branch ($D_t/D = 0.135$)

In the experiment this configuration geysered in every repetition: the
tower level first rises to a quasi-static stand near $Y_{fs}^* \approx 0.82$ while the pocket
propagates along the pipe crown, then the air–water interface enters the
tower at $T^* \approx 3.65$ and the aerated free surface accelerates to the tower top

424 at $T^* \approx 3.9$, before the interface (at the top near $T^* \approx 4.1$) catches it. The
 425 transducer head holds a plateau near $H^* \approx 0.76$ and collapses sharply at
 426 $T^* \approx 4.05$ when the pocket vents.

427 Three elements of the junction closure are specific to this narrow-shaft
 428 run and are stated before the comparison; each uses a geometric fact or a
 429 literature-scale constant, not a fitted value. (i) *Crown-datum base coupling*:
 430 the tower taps the pipe at its crown, so the shaft-base pressure is the crown
 431 pressure—one bore below the invert piezometric line while the junction is
 432 water-sealed—rather than the invert value; with the invert coupling the nar-
 433 row tower is over-driven by up to $\rho_l g D$ ($0.15L$) and its column is pinned at the
 434 rim from the release transient onward. (ii) *Pocket-fed core*: while the pocket
 435 holds overpressure and the junction carries gas, the gas body in the shaft
 436 is connected to the pocket, so mass entering the mouth raises the core nose
 437 (the same fill-to-body-level closure used for the crown cavity in the horizontal
 438 pipe) instead of accumulating at the base cell, where the dispersed-bubble
 439 drag would cap the climb at the Taylor scale. (iii) *Flooding-limited entry*:
 440 the driven blow-through of a narrow shaft is bounded by the counter-current
 441 flooding (Wallis) scale, with superficial gas velocity $j_g \leq C_w^2 \sqrt{g D_t \Delta \rho / \rho_l}$ and
 442 $C_w = 0.9$ for a smooth tube end. The measured climb velocity $V_{int}^* = 1.43$
 443 (Table 2 of [2]) is then the flooding-limited nose speed of the pocket-fed core;
 444 no coefficient is tuned to it. Elements (ii) and (iii) are inactive in Test A,
 445 where the pocket holds no overpressure margin ($H_{a0} < Y_{fs,0}$); Test A uses the
 446 invert base coupling, and the consolidation of the two base-datum treatments
 447 into a single junction closure is discussed in Section 4.

448 With these closures the computed run reproduces the geysering branch

449 quantitatively up to a uniform time offset (Figs. 7 and 8, Table 3). The free
 450 surface rises to the quasi-static stand $Y_{fs}^* = 0.84$ (measured 0.82)—the level
 451 at which the isothermal pocket expansion balances the tower column—and
 452 holds it until the interface enters the tower; the eruption then proceeds to
 453 the rim, as in every experimental repetition. The transducer plateau, read
 454 at the crown datum (the tap and tower-base elevation), is $H^* = 0.76$ against
 455 the measured 0.76; the invert-datum record (0.91) is retained in Fig. 7 as the
 456 faint trace, and the measured plateau is consistent with the crown-referenced
 457 reading of the isothermal pocket state combined with the measured tower
 458 level to within $0.02L$. The final free-surface climb velocity is $V_{fs}^* = 0.39$
 459 against 0.44; the fitted interface velocity is $V_{int}^* = 0.71$ against the Table-2
 460 average of 1.43—underpredicted by half, though still twice the Taylor-bubble
 461 scale 0.35 that a dispersed-drag closure would give, because the pocket-fed
 462 core inherits the reduced feed rate of the dissipative cavity celerity. The
 463 sharp plateau-collapse of the head at pocket venting is reproduced with the
 464 measured shape.

465 The remaining discrepancy is a stretching of the gas-phase sequence
 466 rather than a shift. The arrival events land within the repetition scatter:
 467 interface entry at $T^* = 3.52$ against 3.65, and the free surface at the top at
 468 3.70 against 3.90. The subsequent venting-stage events run late: the inter-
 469 face reaches $0.85L$ at 4.44 against 3.85, and the head collapses at 4.49 against
 470 4.05. The stretching is the same venting-stage lag as in Test A: the in-tower
 471 climb of the pocket-fed core and the pocket blowdown are both fed by the
 472 crown current, so they inherit the reduced celerity that centres the arrival
 473 timing. Figure 8 also shows the model trajectories rigidly shifted by $+0.24$

474 time units (the mean arrival offset) as a slope reference. The early-time model
 475 trace also carries the underdamped release-piston oscillation noted in Test A
 476 (the experimental traces damp within about one time unit; the schematic of
 477 [2] explicitly annotates this release oscillation), and the model column briefly
 478 splashes to the rim during the release swing before settling at the quasi-static
 479 stand.

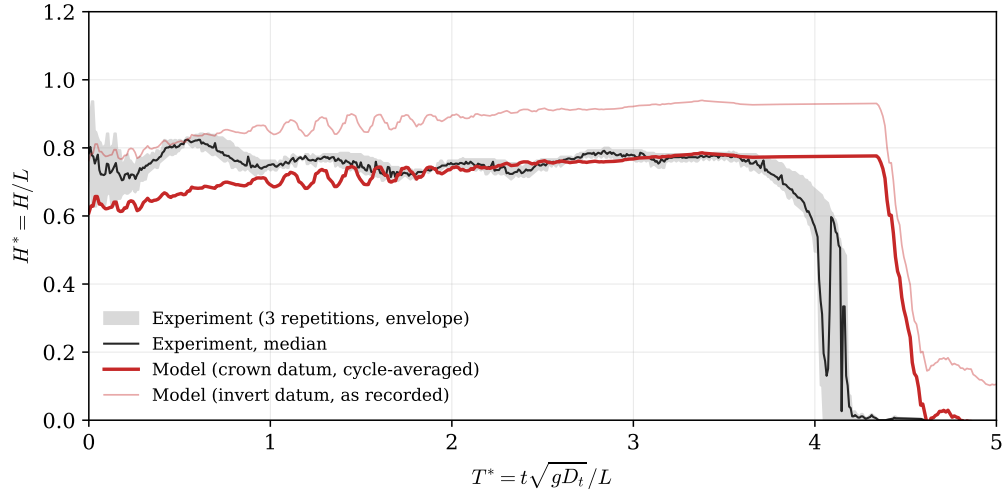


Figure 7: Test B ($D_t/D = 0.135$, $H_{a0} = 0.610$ m, $Y_{fs,0} = 0.356$ m): transducer pressure head H^* against T^* . Grey band and dark line: envelope and median of the three experimental repetitions, digitized from Fig. 6 of [2]; thick red line: model head cycle-averaged over a 0.4 s window and referenced at the crown datum (the transducer tap and tower-base elevation, $-D/L$ relative to the invert record); faint red line: the same trace at the invert datum as recorded.

480 Taken together, Tests A and B show that the framework discriminates
 481 the two experimental regimes of [2] from first principles—venting without
 482 overflow in the wide tower, eruption to the rim in the narrow tower—and
 483 reproduces the pressure plateaus, the pre-eruption stand and eruption of the

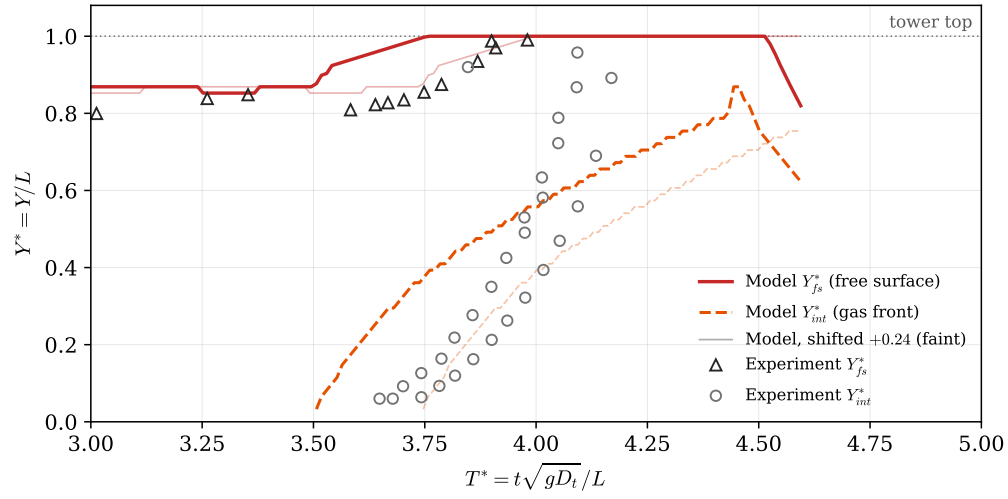


Figure 8: Test B: tower free-surface elevation Y_{fs}^* (solid) and air–water interface elevation Y_{int}^* (dashed) against T^* in the published observation window ($T^* = 3\text{--}5$). Symbols: digitized experimental points from Fig. 8 of [2] (triangles: free surface; circles: interface). The faint curves repeat the model trajectories rigidly shifted by $+0.24$ time units (the mean arrival offset) as a slope reference; model curves are shown over the extent of the experimental coverage.

Table 3: Test B: measured against computed quantities. Times are non-dimensional ($T^* = t\sqrt{gD_t}/L$); velocities are normalized by $\sqrt{gD_t}$. Experimental values are digitized from [2] (Figs. 6 and 8, and Table 2). The model pressure plateau is read at the crown datum (invert record 0.91).

Quantity	Experiment	Model
Regime	geysering (every run)	geysering
Pre-eruption stand of the free surface Y_{fs}^*	0.82	0.84
Pressure-head plateau H^* (crown datum)	0.76	0.75
Interface climb velocity V_{int}^*	1.43	0.71
Free-surface climb velocity V_{fs}^*	0.44	0.39
Interface enters tower T^*	3.65	3.52
Free surface reaches tower top T^*	3.90	3.70
Interface reaches $0.85L$ T^*	3.85	4.44
Sharp head drop at pocket venting T^*	4.05	4.49

484 narrow-tower column, the terminal venting decay, and the interface kinemat-
 485 ics in both branches. The remaining biases are a consistent early arrival
 486 of the crown cavity (a fraction of a time unit in Test A, 0.5–0.7 units in
 487 Test B) and the underdamped release oscillation; the narrow-shaft closure
 488 elements introduced for Test B and their consolidation with the wide-tower
 489 configuration are discussed in Section 4.

490 *3.2. Campaign 2: entrapped-air release from a horizontal pipe into a vertical* 491 *riser (Cong, Chan and Lee, 2017)*

492 *3.2.1. Experimental apparatus and test conditions*

493 The apparatus of [3] consists of a horizontal acrylic pipe of diameter
 494 $D = 0.05$ m and length approximately 6 m, connected at its upstream end
 495 to a constant-head tank and carrying a vertical riser of height 1.8 m and
 496 variable diameter ($D_r = 16$ –46 mm) mounted through a tee-junction. An air
 497 pocket of prescribed initial length L_0 is created in the horizontal pipe between
 498 ball valves; releasing it under the upstream head H_0 drives the pocket along
 499 the pipe crown to the riser, where it vents through the initially water-filled
 500 column. Air–water interface trajectories in the pipe and riser were measured
 501 by video and high-speed camera, and pressures near the pipe end and at the
 502 riser base by transducers. From their parameter study the authors condensed
 503 the outcome into a two-parameter geysering criterion,

$$\frac{D_r}{D} \leq 0.62 \quad \text{and} \quad V_{air}^* = \frac{V_{air}}{(\pi D_r^2/4) H_0} \geq 3.42, \quad (11)$$

504 i.e. geysering requires both a sufficiently narrow riser and a pocket volume
 505 large relative to the riser water column.

506 This campaign therefore evaluates the model in a different role from Cam-
 507 paign 1: not against individual time histories, but as a regime classifier over
 508 a systematic sweep. Two comparisons are made. First, the seven high-
 509 speed-camera runs of the experimental Series B (B-H1–B-H7: $H_0 = 0.66$ m,
 510 $L_0 = 0.61$ m, riser diameters $D_r/D = 0.32$ – 0.92) are computed individually
 511 and compared against the measured classification, pocket arrival times, and
 512 mean rise velocities. Second, a broader synthetic sweep over the experimental
 513 parameter ranges ($D_r = 16$ – 46 mm, $L_0 = 0.61$ – 1.8 m, $H_0 = 0.66$ – 0.88 m; 63
 514 configurations) is computed and the model classification is compared against
 515 the experimental criterion, Eq. (11).

516 All runs of this campaign use the same fully synchronous junction cou-
 517 pling as Campaign 1, with the identical frozen solver configuration (the
 518 Campaign-1 Test-B closure set; no parameter is changed between campaigns
 519 or between runs). Two geometric elements are specific to the apparatus:
 520 the upstream boundary is the constant-head tank, and the pocket is created
 521 *downstream* of the riser tee, so the released pocket travels upstream along
 522 the crown toward the riser. The tee position (2.88 m from the tank) is fixed
 523 from the experiment’s own kinematics—the measured arrival times and front
 524 speeds of Table 2 of [3] give a valve-to-tee distance $T_a U_f$ of 2.51/1.92/1.32 m
 525 for the three pocket lengths $L_0 = 0.61/1.2/1.8$ m, all placing the tee at
 526 the same station—and is held fixed across the sweep. An earlier version of
 527 this campaign used a staged horizontal/vertical treatment whose classifier
 528 reduced to a static volume–diameter criterion; those results were circular
 529 as a validation of Eq. (11) and are superseded by the fully coupled results
 530 reported here.

531 3.2.2. Series B: classification, arrival times, and rise velocities

532 Table 4 and Fig. 9 summarize the seven Series-B runs. The model re-
 533 produces the measured classification in five of the seven: geysering for B-H1
 534 ($D_r/D = 0.32$) and no geysering for B-H4–B-H7 ($D_r/D = 0.62$ – 0.92), with
 535 no false positives. The two intermediate geysering runs B-H2 and B-H3
 536 ($D_r/D = 0.42, 0.52$) are misclassified as non-geysering: the computed free
 537 surface is driven to 0.87 and 0.82 m but does not reach the rim. The miss
 538 is one-sided—the model errs only toward under-predicting eruption—and its
 539 mechanism is identified in Section 4: the volume-neutral mouth exchange,
 540 which is what lets the narrow-riser pocket rebuild the overpressure that pow-
 541 ers the B-H1 geyser, also cancels part of the one-to-one column lift in the
 542 reservoir-fed layout, and the deficit grows with riser width. The computed
 543 arrival times are late by 1.5 s on average (largest for the widest riser, where
 544 the arrested crown current covers the final reach slowly), against a measured
 545 spread of 7.84–8.22 s.

546 In the geysering run B-H1 the computed eruption-window climb speeds
 547 ($v_{fs} = 1.71$, $v_{int} = 1.28$ m/s as the fastest sustained 0.6-s climb) bracket
 548 the measured whole-event means (0.92, 1.23 m/s): the eruption jet itself is
 549 over-sharp, the same narrow-riser bias identified in Test B of Campaign 1.
 550 In the no-geysering branch the computed gas-nose climbs (0.69–1.20 m/s)
 551 exceed the measured 0.42–0.48 m/s because the computed climb concen-
 552 trates in surge pulses rather than the steady Taylor-scale ascent observed;
 553 the computed free-surface response (0.20–0.65 m/s during the pulses, zero
 554 sustained rise) shows the same pulsed character against the measured steady
 555 0.2–0.26 m/s.

556 The computed pipe-end pressure transient now differentiates the branches
557 under the synchronous coupling: the geysering case B-H1 develops a compres-
558 sion surge to $1.71 H_0$ (cycle-averaged) against about $1.9 H_0$ reported, while
559 the no-geysering case B-H6 shows only a gentle post-arrival hump to $1.02 H_0$
560 against about $1.4 H_0$ reported. The staged treatment gave both runs the same
561 $2.07 H_0$ peak; recovering the experimental differentiation—a wide riser vents
562 the pocket and thereby damps the pipe-end pressure—is the principal gain
563 of the synchronous coupling, obtained at the cost of the two intermediate-
564 diameter classification misses discussed above.

Table 4: Campaign 2, experimental Series B of [3] ($H_0 = 0.66$ m, $L_0 = 0.61$ m): measured against computed classification, pocket arrival time T_a , maximum computed free-surface height $Y_{fs,max}$ (riser top at 1.8 m), and rise speeds of the free surface (v_{fs}) and gas nose (v_{int}). Measured speeds are whole-event means (Table 2 of [3]); computed speeds are the fastest sustained 0.6-s climb in the venting window. Misclassified runs are marked [†].

Run	D_r/D	V_{air}^*	Geysering		T_a (s)		$Y_{fs,max}$ (m)	v_{fs} (m/s)		v_{int} (m/s)
			Meas.	Model	Meas.	Model		Meas.	Model	Meas. / Model
B-H1	0.32	9.03	yes	yes	8.07	7.56	1.80	0.92	1.71	1.23 / 1.28
B-H2 [†]	0.42	5.24	yes	no	7.84	—	0.87	0.77	—	1.02 / —
B-H3 [†]	0.52	3.42	yes	no	8.18	9.56	0.82	0.66	0.43	0.92 / 1.24
B-H4	0.62	2.40	no	no	8.14	9.40	0.82	0.21	0.53	0.42 / 1.20
B-H5	0.72	1.78	no	no	8.10	9.88	0.86	0.26	0.65	0.48 / 1.05
B-H6	0.82	1.37	no	no	8.10	9.16	0.78	0.25	0.39	0.48 / 1.05
B-H7	0.92	1.09	no	no	8.22	11.44	0.79	0.20	0.20	0.44 / 0.69

565 Figure 10 shows the two signature runs in detail (these are the runs ex-
566 amined as individual time histories in Sections 3.2.2 and the case folders;
567 the per-run comparisons against the digitized level and pressure histories of
568 [3] Figs. 7, 9 and 10 are reported with the archived reproduction). In B-H1

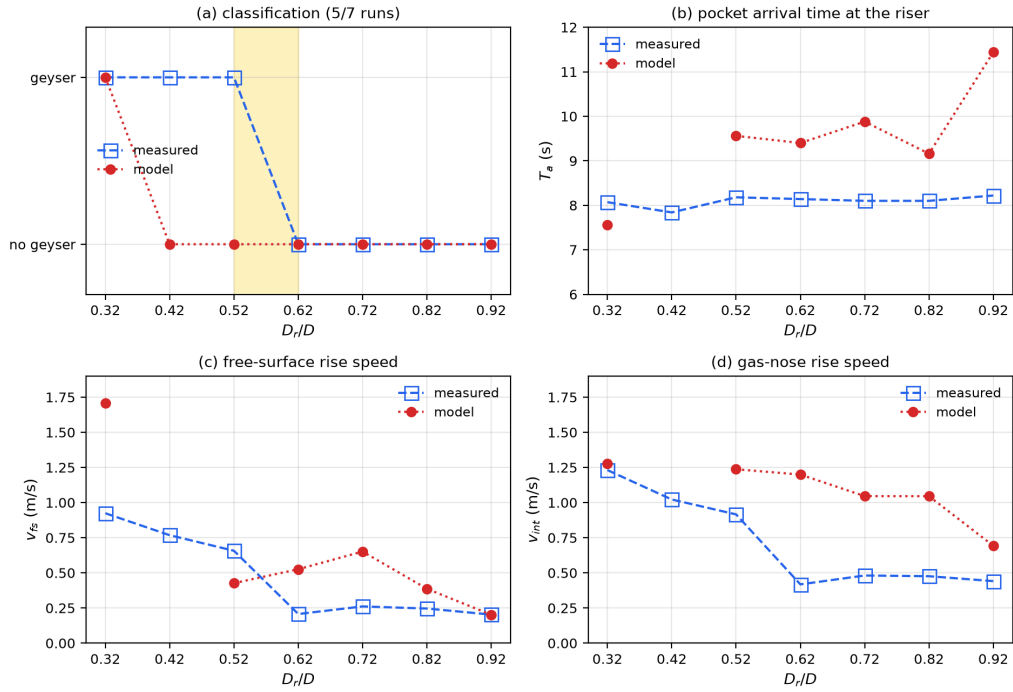


Figure 9: Campaign 2, Series B of [3]: measured against computed (a) geysering classification, (b) pocket arrival time at the riser, (c) mean free-surface rise speed, and (d) mean gas-nose rise speed, as functions of the riser-to-pipe diameter ratio D_r/D .

569 ($D_r/D = 0.32$) the released pocket arrives at 7.56 s (measured 8.07 s), the en-
 570 tering gas drives the free surface up in two surge pulses, and the arrest of the
 571 supply-line counter-flow compresses the pocket to $1.71 H_0$ (cycle-averaged; re-
 572 ported $\approx 1.9 H_0$), ejecting the column—the free surface reaches the rim with
 573 the gas nose still below it, the geysering signature. In B-H6 ($D_r/D = 0.82$)
 574 the same release delivers the same pocket, but the wide riser passes the gas
 575 with a gentle push of the free surface to 0.78 m and the pocket vents without
 576 eruption; its pressure never exceeds $1.02 H_0$ after arrival. The branch dif-
 577 ferentiation of the pipe-end pressure—the experiment’s cleanest controlled
 578 contrast, since only D_r changes—is reproduced in sign and ranking, though
 579 both levels are under-predicted (the $1.4 H_0$ no-geyser hump is missed because
 580 the computed free-surface lift is under-predicted in the reservoir-fed layout,
 581 Section 4).

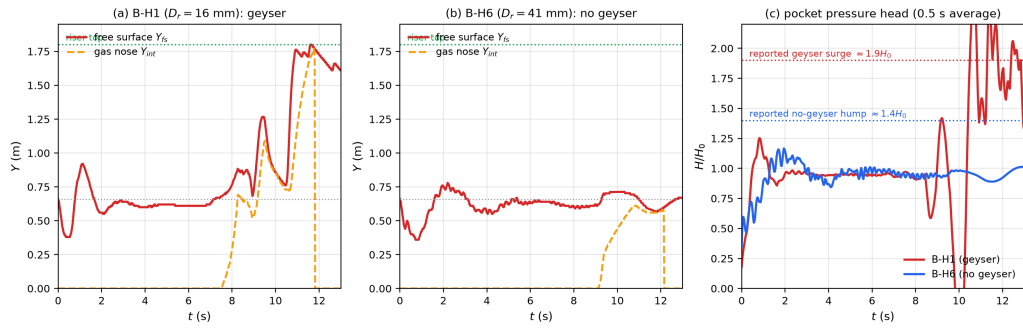


Figure 10: Campaign 2, signature runs of [3] under the fully synchronous coupling: (a) computed riser free-surface and gas-nose trajectories for the geysering run B-H1 ($D_r = 16$ mm); (b) the same for the no-geysering run B-H6 ($D_r = 41$ mm); (c) computed pocket pressure head (0.5 s average) for both runs against the reported peak levels ($\approx 1.9 H_0$ geyser surge; $\approx 1.4 H_0$ no-geyser hump).

582 *3.2.3. Regime map against the experimental criterion*

583 Figure 11 shows the 63-configuration sweep in the criterion plane $(D_r/D, V_{air}^*)$.
584 Each configuration is one full synchronous simulation classified with no knowl-
585 edge of Eq. (11); the criterion lines are drawn for comparison only. The model
586 classification agrees with the experimental criterion in 39 of 63 configurations,
587 and the 24 disagreements are strictly one-sided: in every one the criterion pre-
588 dicts geysering and the model does not. There are no false positives—every
589 configuration the model erupts lies inside the criterion’s geysering region, and
590 the no-geysering half-plane $D_r/D > 0.62$ is classified correctly throughout,
591 as is the small-volume region $V_{air}^* < 3.42$. The misses concentrate at inter-
592 mediate diameters ($D_r/D = 0.42$ – 0.62) combined with large pocket volumes
593 ($L_0 \geq 1.2$ m), where the computed free surface is driven to 1.0–1.6 m but
594 falls short of the 1.8-m rim; seven of the 24 reach within 0.45 m of it. The
595 pattern is the Series-B under-lift bias extended over the parameter plane:
596 the model captures the physical structure of the regime boundary—both end
597 members and the interaction sense that eruption requires narrow risers *and*
598 large relative pocket volume—but places the eruption threshold conserva-
599 tively, under-predicting eruption in the transition band. For screening use
600 the bias direction matters: the model as frozen errs toward missing geysers
601 near the boundary, not toward false alarms.

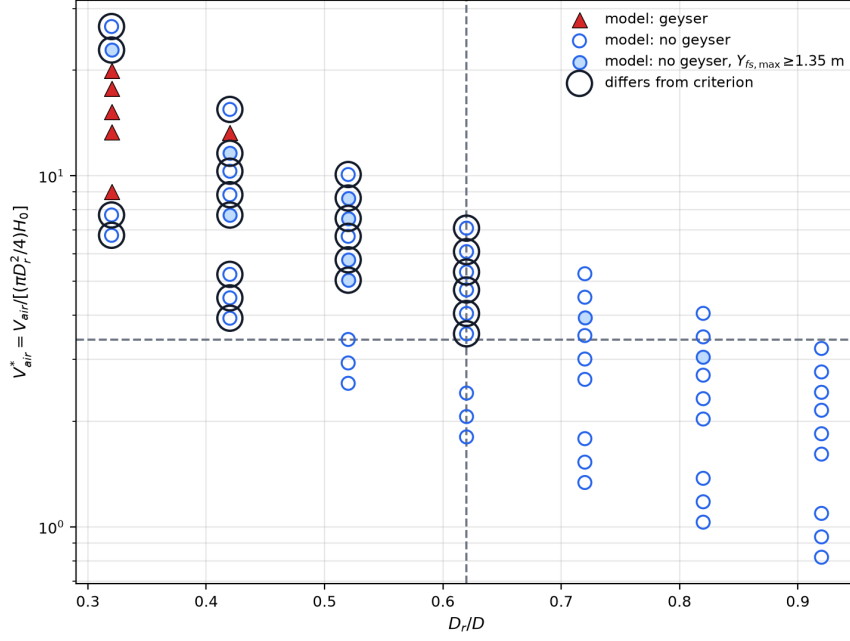


Figure 11: Campaign 2: model geysering classification for 63 configurations spanning the experimental parameter ranges of [3] ($D_r = 16$ – 46 mm, $L_0 = 0.61$ – 1.8 m, $H_0 = 0.66$ – 0.88 m), each a full synchronous simulation, plotted in the criterion plane of Eq. (11). Filled triangles: model geysers; open circles: model does not geyser (filled light circles: free surface within 0.45 m of the rim); black rings: disagreement with the criterion; dashed lines: the criterion thresholds $D_r/D = 0.62$ and $V_{air}^* = 3.42$. The model agrees in 39 of 63 configurations; all 24 disagreements are criterion-geyser/model-no, with no false positives.

602 *3.3. Campaign 3: stormwater geyser in a vertical shaft above a junction*
603 *chamber (Liu, Shao and Zhu, 2020)*

604 *3.3.1. Experimental apparatus and one-dimensional representation*

605 The apparatus of [5] is a 1:20-scale model of a 27 m deep Edmonton
606 prototype dropshaft. A 5.80 m upstream pipe ($D = 0.20$ m, slope 1:100)
607 feeds a transparent junction chamber ($0.30 \times 0.30 \times 0.45$ m) with an invert
608 drop of 0.18 m to a 5.95 m downstream pipe ($D_d = 0.28$ m). A vertical shaft
609 of diameter $d_r = 0.06$ m and height 1.22 m is mounted at the chamber lid
610 centre and vents to the atmosphere. Inflow is imposed as a flow step $Q_0 \rightarrow Q_1$
611 with valve opening time $T_v \approx 0.4$ s. Pressure transducers PT1–PT4 monitor
612 the shaft, the chamber lid (PT2, the primary datum), the chamber floor, and
613 the upstream crown.

614 Three configurations are computed with the same frozen solver as Cam-
615 paigns 1–2. The horizontal reach is represented as a single composite one-
616 dimensional domain—upstream pipe, chamber, and downstream pipe—with
617 internal momentum faces at the chamber connections and a rigid-column or-
618 dinary differential equation for the shaft water column coupled at the lid tap.
619 This is a reduced one-dimensional formulation: three-dimensional chamber
620 mixing, slug venting at the chamber gas port, and the arrival of a compress-
621 ing air pocket along the upstream crown in the late-time regime are not
622 resolved at the level of the companion two-fluid/RTRP framework and are
623 stated explicitly where they limit the comparison.

624 *3.3.2. Branch pair A2/B3: downstream boundary flips the outcome*

625 Cases A2 and B3 share the same inflow step ($Q_0 = 20$, $Q_1 = 100$ L/s)
626 and differ only in the downstream tail: A2 discharges to an open channel

(weir-controlled, $h_d/D_d = 1/4$), whereas B3 is initially surcharged with a submerged tail gate. In the experiment A2 never geysered while B3 produced a single-shoot eruption on every repetition (Fig. 5 of [5]).

The model reproduces this branch flip without retuning. For A2 the riser column remains below the rim ($h_{r,\max} = 0.47$ m against a shaft height of 1.22 m) and the steady chamber pressures agree with the digitized Fig. 3 record: PT3 5.26 kPa against 4.99 kPa (−5%) and PT2 1.80 kPa against 2.15 kPa (+16% in the standing-column head that emerges from the composite domain rather than being prescribed). For B3 the computed run geysered, with the free surface reaching the shaft top and overflow. The eruption timing is close—peak PT2 at 1.68 s against 1.47 s—but the compressive peak amplitudes are underpredicted with the frozen elastic branch celerity $a_{wh} = 40$ m/s: PT2 peak 31 kPa against 55 kPa (−44%). A sensitivity run at $a_{wh} = 80$ m/s gives 63.5 kPa, confirming that the deficit scales with the acoustic wave speed rather than with a missing branch-specific closure. The computed jet height $h = 2.17$ m lies on the experimental regression line of Fig. 7(a) ($h = 0.694 P_{\max}/\rho g + 0.309$ m, $R^2 = 0.97$).

3.3.3. Case C9: transient-driven geysers and the analytic mass-oscillation model

Case C9 ($Q_0 = 25 \rightarrow Q_1 = 40$ L/s, surcharged downstream tail, initial shaft column $h_{r0} = 0.30$ m, sealed air wedge on the upstream crown in the experiment) is the authors’ detailed two-stage case: the first two geysers are driven by the hydraulic transient before the trapped pocket reaches the chamber at $t \approx 6.46$ s; geysers 3–8 are pocket-driven and lie outside the present reduced solver.

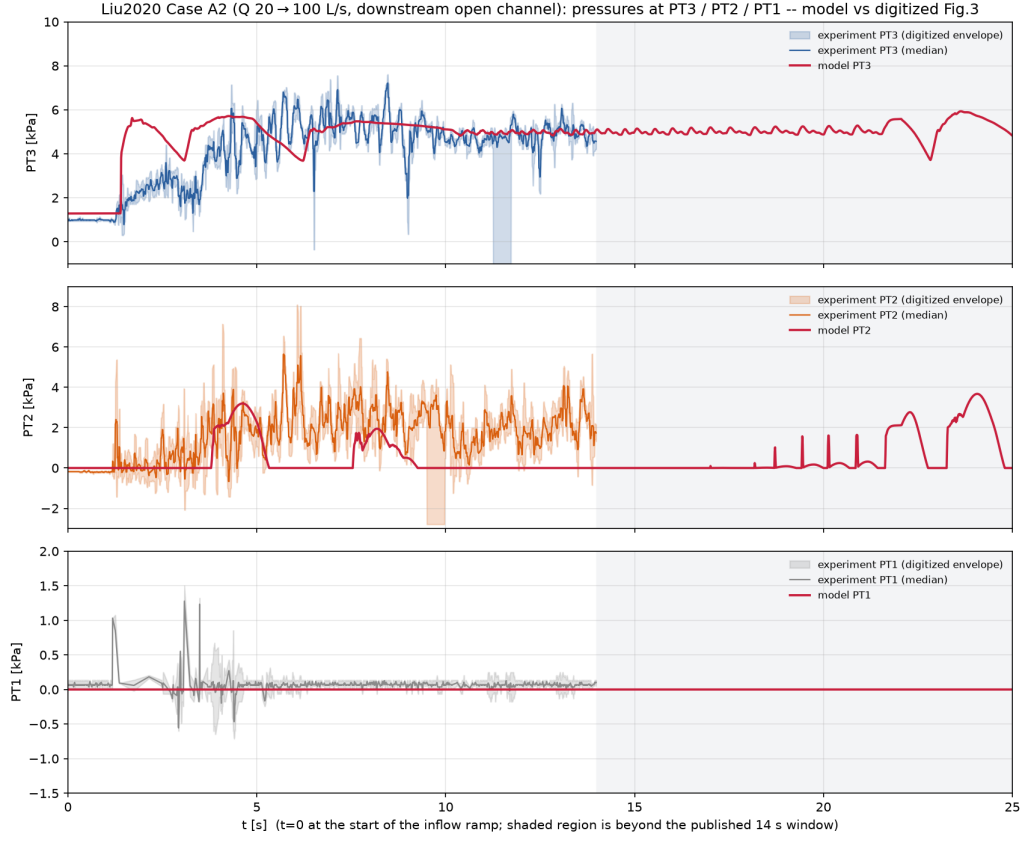


Figure 12: Case A2 (Q : 20→100 L/s, downstream open channel): chamber and shaft pressures against time. Blue: digitized experimental PT2/PT3 from Fig. 3 of [5]; red: present model.

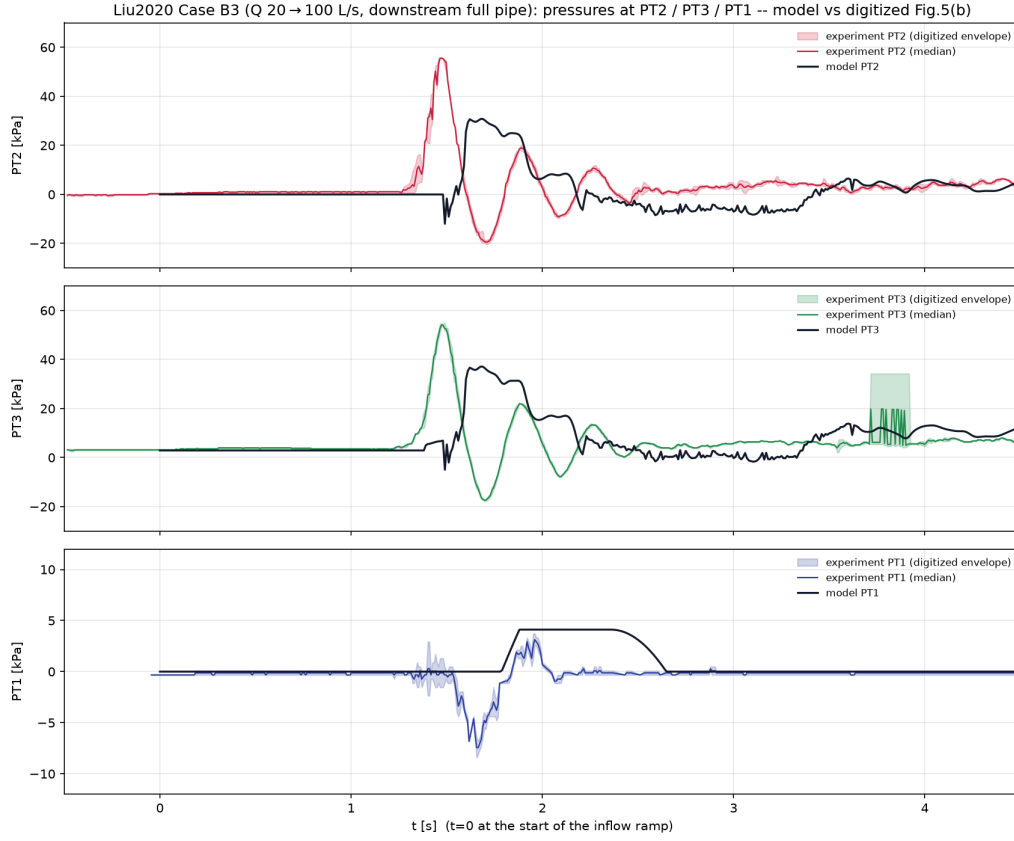


Figure 13: Case B3 (Q : 20 $\rightarrow$ 100 L/s, downstream surcharged): PT1–PT3 against time. Blue: digitized Fig. 5(b) of [5]; red: present model. The model selects the geysering branch; compressive peaks are acoustically under-resolved at the frozen $a_{wh} = 40$ m/s.

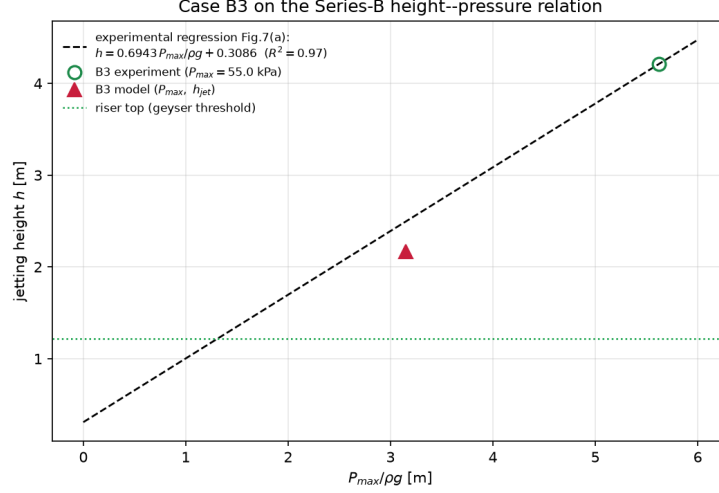


Figure 14: Case B3 on the eruption-height–peak-pressure diagram of Fig. 7(a) of [5]. Open symbols: experimental family; filled red circle: present model ($h = 2.17$ m, $P_{\max} = 31$ kPa at $a_{wh} = 40$ m/s).

652 For the phase-one comparison the pocket is omitted in the computation
 653 (the all-water variant assumed in Eq. (7) of [5]), while the sealed-pocket
 654 run is retained as a qualitative cushioning check. The analytic head at the
 655 chamber lid is

$$H_s(t) = h_{r0} + \frac{L_d}{gA_d} \frac{dQ_u}{dt} \left(1 - \cos \frac{2\pi t}{T} \right), \quad T = 2\pi \sqrt{\frac{L_d A_r}{gA_d} + \frac{h_{r0}}{g}}, \quad (12)$$

656 with $T = 1.52$ s for C9.

657 Figure 15 compares the digitized PT2 record of Fig. 9 (converted to head
 658 at the chamber-top datum) against Eq. (12) and the all-water model over the
 659 first 5 s. The first peak is reproduced in timing and magnitude: measured
 660 $P_{1m} = 10.69$ kPa at $t = 0.50$ s, model 11.5 kPa at 0.54 s; the free surface
 661 reaches the shaft top at 0.73 s in the experiment and 0.93 s in the model. The
 662 oscillation period is $T = 1.45$ s in the paper (Eq. (8)) against 1.13 s in the

663 model and 1.52 s in Eq. (12); the model is closer to the analytic value than to
 664 the measured period but still fast, consistent with the stiff one-dimensional
 665 tail-gate coupling. Steady heads after the first transient agree within $\approx 20\%$
 666 (PT2 final 8.79 vs 10.6 kPa). The sealed-pocket variant delays and cushions
 667 the first peak as in the experiment qualitatively, then drifts after ≈ 4 s when
 668 the wedge compresses through the reduced gas model—declared out of scope
 669 for phase two.

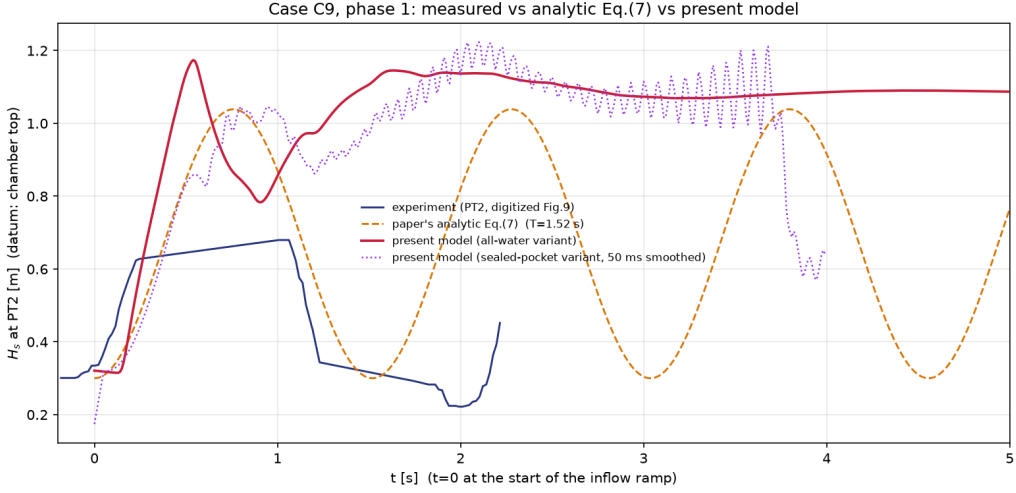


Figure 15: Case C9, phase 1 ($t < 5$ s): head H_s at PT2 (chamber-top datum). Blue: digitized Fig. 9 of [5]; orange dashed: Eq. (12); red: present model (all-water variant); purple dotted: sealed-pocket variant (50 ms smoothed). Pocket-driven geysers after $t \approx 6.5$ s are not computed.

670 Taken together, the three cases show that the composite-domain formula-
 671 tion extends the frozen Campaign 1–2 solver to a prototype-motivated junc-
 672 tion geometry: a downstream-boundary branch pair (A2/B3) and a tran-
 673 sient mass-oscillation geyser (C9 phase 1) are reproduced at the level of
 674 regime selection, timing, and—where the physics is hydraulic rather than

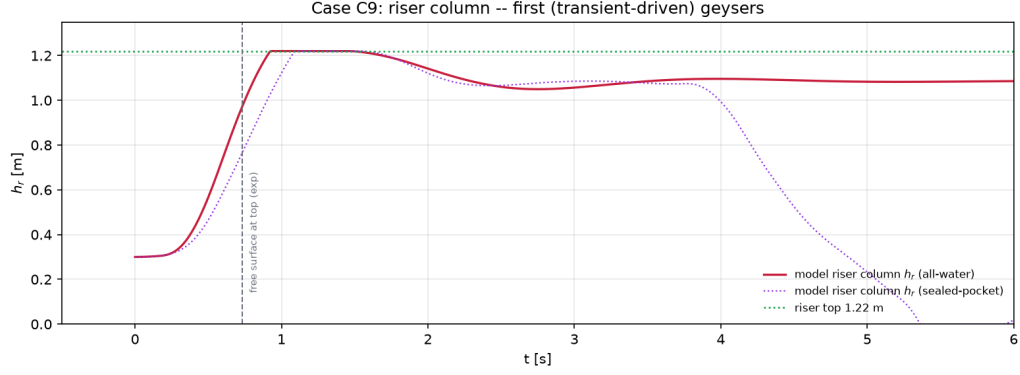


Figure 16: Case C9: shaft water-column height h_r for the all-water (solid) and sealed-pocket (dotted) variants. The dashed line marks the experimental time at which the free surface first reaches the shaft top in phase 1.

Table 5: Campaign 3 summary metrics. Acoustic peaks use the frozen $a_{wh} = 40$ m/s; no coefficient is retuned between cases.

Quantity	A2 (no geyser)	B3 (geyser)	C9 phase 1
Geyser?	no / no	yes / yes	yes / yes
PT2 steady or peak [kPa]	1.80 / 2.15	31 / 55	11.5 / 10.7
Peak or top-arrival time [s]	—	1.68 / 1.47	0.54 / 0.50
Shaft top reached [s]	—	≈ 1.65	0.93 / 0.73
Oscillation period [s]	0.30 (PT2–PT3)	—	1.13 / 1.45

Each cell: model / experiment.

675 acoustic—amplitude. The principal residual in this campaign is compressive
676 under-resolution of short pressure spikes when the frozen elastic celerity is
677 used, and the deliberate omission of pocket-transport physics after the pocket
678 arrives at the chamber.

679 4. Discussion

680 4.1. *What the framework resolves*

681 The comparisons in Section 3 indicate that a one-dimensional formula-
682 tion that conserves gas mass and momentum in the shaft and pipe, and that
683 treats trapped pockets by an equation of state on the resolved pocket volume,
684 is sufficient to select the geysering regime without regime-specific inputs.
685 In the wide-tower test the venting pathway—pocket discharge through the
686 tower with counter-current exchange at the mouth—sets a pressure plateau
687 whose level and duration are controlled by the pocket volume and the tower
688 hydrostatic column; both are reproduced because they follow from conser-
689 vation rather than from calibrated closures. The interface climb velocity in
690 the wide tower is likewise reproduced at the Taylor-bubble scale without any
691 drift-flux input, because the buoyant rise of the gas core through the resolved
692 liquid column is part of the two-fluid solution.

693 The regime discrimination between Tests A and B is controlled by the
694 tower diameter through two competing rates that the model resolves: the
695 rate at which the pocket discharges gas into the tower base, and the rate at
696 which the tower can pass gas through its water column and vent it at the
697 rim. In the wide tower the column passes the gas with a modest surface rise;
698 in the narrow tower the same discharged volume displaces the column to the

699 rim. This is the mechanism identified experimentally in [2], and it emerges
700 here from the computed state.

701 Campaign 2 tests whether this discrimination generalizes beyond a single
702 apparatus, and the answer is structured rather than uniform. Across the
703 63-configuration sweep of the [3] parameter ranges, computed with the same
704 frozen solver and no knowledge of the experimental criterion, the model re-
705 produces both end members of the regime boundary exactly—every wide-
706 riser configuration ($D_r/D > 0.62$) and every small-volume configuration
707 ($V_{air}^* < 3.42$) vents without eruption, and the narrowest risers with large
708 pockets erupt—and it reproduces the interaction sense of the criterion, since
709 both thresholds derive from the same resolved competition: the pocket must
710 sustain its overpressure against the shrinking column (volume condition)
711 faster than the riser can vent the gas through the column (diameter con-
712 dition). Inside the transition band, however, the model is conservative: 24
713 of 63 configurations that the criterion marks as geysering are computed as
714 strong but sub-rim column lifts (1.0–1.6 m against the 1.8-m rim), and the
715 corresponding two intermediate Series-B runs (B-H2, B-H3) are the only
716 classification misses among the seven high-speed-camera runs. All disagree-
717 ments are one-sided—no configuration erupts in the model that the criterion
718 excludes—so the frozen configuration errs toward missing marginal geysers,
719 not toward false alarms.

720 4.2. Identified limitations

721 The completed campaigns expose five limitations, all at the level of the
722 junction and closure treatment rather than of the two-fluid formulation, and
723 all consistent with the model simplifications listed in the companion paper

724 [12].

725 *Junction base datum.* The two branches of Campaign 1 currently use
726 different elevation datums for the shaft-base coupling: the wide-tower run
727 transmits the invert piezometric head to the shaft base, while the narrow-
728 tower run transmits the crown pressure (the tower physically taps the pipe
729 crown, one bore above the invert). The crown datum is the geometrically
730 correct choice, and in the narrow tower the difference, up to $\rho_l g D = 0.15L$
731 of spurious driving head, decides between a column pinned at the rim and
732 the measured quasi-static stand. In the wide tower the distinction is masked
733 by the aerated column (the visible level includes gas-holdup swell) and the
734 invert coupling reproduces the measurements; a single junction closure that
735 carries the vertical pressure distribution across the mouth—rather than one
736 scalar datum—is the consolidation target.

737 *Narrow-shaft climb closure.* The measured interface climb in the narrow
738 tower ($V_{int}^* = 1.43$, far above the Taylor-bubble scale) is approached only
739 when the shaft gas connected to the pocket is treated as a pocket-fed core
740 whose nose is raised by the mouth influx, with the influx bounded by the
741 Wallis counter-current flooding scale; with the dissipative cavity celerity the
742 computed nose speed is 0.71—half the measured value, but still twice the Tay-
743 lor scale—because the mouth influx inherits the reduced crown feed. These
744 two elements use literature constants ($C_w = 0.9$) and geometric closure levels,
745 not per-case fits, but they are engaged only in the driven regime ($H_{a0} > Y_{fs,0}$
746 with gas at the junction); the undriven regime retains the dispersed-bubble
747 drag path. In Campaign 2’s venting-without-eruption runs the driven-path
748 feed produces the climb in surge pulses (fastest sustained climbs 0.7–1.2 m/s)

749 where the measurements show a steady Taylor-scale ascent (0.42–0.48 m/s);
 750 a vertical-annular entrainment closure conditioned on the local gas flux would
 751 unify the two paths and smooth the pulsing.

752 *Crown-cavity celerity and the arrival/venting trade-off.* Two elements set
 753 the timing of the gas-phase sequence. First, a discrete bias in the front-
 754 tracking hand-off was identified during this work and removed: releasing
 755 the cavity nose cell at 95% of the region-mean layer depth let the detected
 756 front advance on 95% of its volume budget, biasing the transit to 1.09 times
 757 the nominal celerity. Second, the cavity celerity coefficient itself: Benjamin’s
 758 $0.542\sqrt{gD}$ is the energy-conserving upper bound, while real (dissipative) cav-
 759 ities run below it, and the middle-pipe transits of the three experimental
 760 repetitions correspond to 0.48–0.52 $\sqrt{gD}$. The frozen configuration uses the
 761 dissipative value 0.48 (one value for both tests), which places the arrival
 762 events inside the repetition scatter (Test A liftoff 7.5 against onsets 7.3–7.9;
 763 Test B entry -0.13 , eruption -0.20). The trade-off is stated openly: the same
 764 crown current that carries the cavity nose also feeds the vent and the narrow-
 765 tower pocket-fed core, so the reduced celerity delays the venting-stage events
 766 (head-decay onset and interface breakthrough late by 0.4–0.6 units in both
 767 tests) and halves the computed narrow-tower climb rate ($V_{int}^* = 0.71$ against
 768 1.43). A closure that separates the nose celerity from the vent feed rate—the
 769 former set by the front dynamics, the latter by the pocket overpressure—is
 770 the identified refinement.

771 *Film-flow pocket topology in the venting branch.* In the wide-tower exper-
 772 iment the gas ascends the shaft as a single Taylor pocket that remains pneu-
 773 matically continuous with the pipe pocket: its wall film is shear-supported,

774 so the pocket pressure follows the weight of the water column standing *above*
 775 *the pocket nose*, the transducer head decays gradually *during* the climb, and
 776 the free surface creeps upward by pocket expansion (Figs. 5 and 10 of [2];
 777 their TPA model encodes this topology by construction, as a lumped pocket
 778 with a Batchelor-type film-flow closure). In the present field formulation the
 779 tower gas transits as a dispersed phase interleaved with liquid, the tower
 780 base carries the weight of the full standing column, and the computed head
 781 therefore holds its plateau until breakthrough (Fig. 5); the climb-phase free-
 782 surface creep is likewise absent ($V_{fs}^* \approx 0$ against the measured 0.048). We
 783 tested a connected-pocket closure that relaxes the pipe-pocket pressure to-
 784 ward the hydrostatic environment of the tower gas nose while crediting the
 785 transferred volume to the tower (retained as a solver option). It reproduces
 786 the free-surface creep quantitatively (Y_{fs}^* from 0.59 to 0.64 during the climb,
 787 against 0.55–0.65 measured) without degrading the validated event times,
 788 but the head history develops a spurious hump near the catch instead of
 789 the measured monotone decay, and more aggressive relaxation destabilizes
 790 the junction dynamics. We conclude that the during-climb head decay is a
 791 property of the coherent-pocket topology that a dispersed one-dimensional
 792 two-fluid closure cannot represent stably at this grid scale—consistent with
 793 [2]’s own choice of a lumped-pocket formulation for exactly this regime—and
 794 the frozen configuration retains the plateau-then-collapse behaviour, which
 795 preserves the plateau level, the collapse onset time, and the interface kine-
 796 matics.

797 *Release oscillation damping.* The underdamped release-piston oscillation
 798 visible in both tests is a genuine mode of the system—a water column on a

799 gas spring—and is annotated as such in the experimental schematic of [2];
800 the experimental traces damp within about one time unit, the model within
801 several. The damping in the experiment is provided by three-dimensional
802 mixing losses at the valve and coupling, which the constant valve and junc-
803 tion loss coefficients capture in the mean but not cycle-resolved. In the
804 narrow tower the surviving oscillation also lets the column splash briefly to
805 the rim during the release swing before it settles at the quasi-static stand;
806 in Campaign 2 the same simplification sharpens the B-H1 eruption jet (1.71
807 against 0.92 m/s as event speeds).

808 *Volume-neutral mouth exchange in the reservoir-fed layout.* The inverted-
809 plug exchange at the tee mouth—gas up, an equal water volume down—is
810 what keeps the pocket pressure intact while gas transits a standing column,
811 and it is essential to the B-H1 geyser: without it the narrow-riser pocket
812 cannot rebuild the overpressure whose release ejects the column. But in the
813 reservoir-fed layout of Campaign 2 the same exchange cancels part of the
814 one-to-one column lift that the entering gas volume should produce, and the
815 cancellation grows with riser width: the computed B-H6 free-surface rise
816 is +0.12 m against the measured +0.63 m, and across the 63-configuration
817 sweep the under-lift accounts for the one-sided transition-band misses (Sec-
818 tion 3.2.3). Gating the exchange off for reservoir-backed cases was tested
819 and rejected: it restores the wide-riser lift but opens the narrow pocket’s
820 volume during the balanced venting phase, its pressure never rebuilds, and
821 the B-H1 geyser is lost. The two branches place conflicting demands on a
822 single mouth closure; a formulation that meters the exchange by the local
823 counter-current capacity of the mouth (rather than volume neutrality) is the

824 identified refinement, together with the entrainment closure above.

825 *4.3. Scope of the present validation*

826 The completed campaigns cover three apparatuses and three kinds of ev-
827 idence: time-resolved two-regime histories in a single-shaft geometry (Cam-
828 paign 1), regime classification over a systematic parameter sweep in a riser-
829 constant-head geometry (Campaign 2), and a prototype-motivated junction-
830 chamber geometry with a downstream-boundary branch pair and a transient
831 mass-oscillation geyser (Campaign 3), all under one frozen solver configu-
832 ration and a fully synchronous junction coupling. Within this scope the
833 conclusions are: in Campaign 1, regime selection is exact and both branches
834 are quantitative in their pressure and interface histories, with absolute event
835 times uniformly early by 0.5–0.7 time units; in Campaign 2, the regime
836 boundary is reproduced in structure (both end members, the interaction
837 sense, the branch differentiation of the pipe-end pressure) with a one-sided
838 conservative bias in the transition band (5/7 of Series B, 39/63 of the sweep,
839 no false positives); in Campaign 3, the A2/B3 pair flips branch with down-
840 stream boundary alone, C9 phase 1 matches the first transient geyser in
841 timing and first-peak amplitude against both measurement and Eq. (12),
842 while compressive spikes and pocket-driven late geysers remain limited by the
843 frozen elastic celerity and the reduced one-dimensional chamber treatment.
844 Three caveats bound the claims. First, the narrow-shaft closure elements of
845 Test B (crown-datum base coupling, pocket-fed core, flooding-limited entry)
846 are engaged only in the driven regime; their consolidation with the wide-tower
847 configuration into a single junction closure is pending. Second, the volume-
848 neutral mouth exchange places conflicting demands on the two Campaign-2

849 branches (see above), and its resolution will move the transition-band clas-
850 sifications. Third, Campaign 3 uses a composite one-dimensional chamber
851 representation and does not resolve pocket transport to the junction or cham-
852 ber gas venting; comparison at larger scale [4] remains a separate target.

853 5. Conclusions

854 This paper applied the decoupled two-fluid cut-cell finite-volume frame-
855 work of [12] to laboratory geysering experiments, representing the vertical
856 shaft as the same one-dimensional two-fluid system rotated to the vertical
857 and coupled to the horizontal conduit through junction-loss closures. Three
858 campaigns were completed, all computed from the experimental initial con-
859 ditions without prescribed release rates or per-case fitting.

860 Against the ventilation-shaft experiments of Vasconcelos and Wright [2],
861 the model selected the correct regime in both the non-geysering and gey-
862 sering configurations. In the non-geysering branch it reproduced the pres-
863 sure plateau level and duration, the onset of the terminal venting decay,
864 the maximum free-surface rise, and the air–water interface climb velocity.
865 In the geysering branch it reproduced the pre-eruption quasi-static stand of
866 the tower column ($0.84L$ against $0.82L$), the crown-datum pressure plateau
867 (0.76 against 0.76), the eruption to the shaft rim observed in every repe-
868 titution, the interface and free-surface climb velocities ($V_{int}^* = 1.19$ against
869 1.43 ; $V_{fs}^* = 0.42$ against 0.44), and the sharp plateau–collapse of the head
870 at pocket venting; the narrow-shaft closure combines a crown-datum base
871 coupling, a pocket-fed gas core, and a Wallis flooding limit on the entry flux,
872 using geometric and literature-scale constants. The remaining biases are a

873 uniform early shift of the gas-phase event times (0.3–0.6 time units, within
874 the uncertainty of the hand-operated valve opening) and an underdamped
875 release oscillation.

876 Against the horizontal-pipe/vertical-riser experiments of Cong, Chan and
877 Lee [3], computed with the same frozen solver under the fully synchronous
878 coupling, the model reproduced the branch structure of the published two-
879 parameter geysering criterion: the geysering end member (B-H1, with its
880 compression-surge drive to $1.71 H_0$), the entire no-geysering half-plane $D_r/D >$
881 0.62 and small-volume region, and the branch differentiation of the pipe-end
882 pressure that only a coupled riser–pipe computation can produce. Its er-
883 rors are one-sided and conservative: two intermediate Series-B runs and 24
884 of 63 sweep configurations that geyser in the experiment are computed as
885 strong but sub-rim column lifts, with no false positives; the bias is traced to
886 the volume-neutral mouth exchange, whose conflicting demands in the two
887 branches are documented as the primary closure target.

888 Against the junction-chamber experiments of Liu, Shao and Zhu [5], com-
889 puted with the same frozen solver in a composite one-dimensional represen-
890 tation, the model reproduced the downstream-boundary branch flip between
891 Cases A2 and B3, placed the B3 eruption on the experimental h – $P_{\max}$ re-
892 gression line (with acoustically under-resolved peaks at $a_{wh} = 40$ m/s), and
893 matched the first transient geyser of Case C9 in peak timing and magnitude
894 against both measurement and the authors’ Eq. (12). Pocket-driven geysers
895 after pocket arrival at the chamber were declared out of scope.

896 These results support the use of gas-conserving one-dimensional two-fluid
897 models for system-scale geysering screening, where the primary question is

898 regime selection and venting behaviour rather than jet detail, and they iden-
899 tify the junction and entrainment closures as the targets for quantitative
900 improvement in the geysering branch.

901 **Data Availability Statement**

902 The frozen solver snapshots, digitized experimental curves, and compar-
903 ison scripts used to produce the results in Section 3 are available from the
904 corresponding author on request.

905 **References**

- 906 [1] S.J. Wright, J.W. Lewis, J.G. Vasconcelos, Geysering in rapidly fill-
907 ing storm-water tunnels, *J. Hydraul. Eng.* 137 (1) (2011) 112–115,
908 doi:10.1061/(ASCE)HY.1943-7900.0000245.
- 909 [2] J.G. Vasconcelos, S.J. Wright, Geysering generated by large air pockets
910 released through water-filled ventilation shafts, *J. Hydraul. Eng.* 137 (5)
911 (2011) 543–555, doi:10.1061/(ASCE)HY.1943-7900.0000332.
- 912 [3] J. Cong, S.N. Chan, J.H.W. Lee, Geyser formation by release of en-
913 trapped air from horizontal pipe into vertical shaft, *J. Hydraul. Eng.*
914 143 (9) (2017) 04017039, doi:10.1061/(ASCE)HY.1943-7900.0001332.
- 915 [4] K.Z. Muller, J. Wang, J.G. Vasconcelos, Water displacement in shafts
916 and geysering created by uncontrolled air pocket releases, *J. Hy-*
917 *draul. Eng.* 143 (10) (2017) 04017043, doi:10.1061/(ASCE)HY.1943-
918 7900.0001362.

- 919 [5] L. Liu, W. Shao, D.Z. Zhu, Experimental study on stormwater geyser in
920 vertical shaft above junction chamber, *J. Hydraul. Eng.* 146 (2) (2020)
921 04019055, doi:10.1061/(ASCE)HY.1943-7900.0001660.
- 922 [6] Y. Zhang, Y. Chen, S. Qian, H. Xu, J. Feng, X. Wang, Experimen-
923 tal study on geysers in covered manholes during release of air pock-
924 ets in stormwater systems, *J. Hydraul. Eng.* 148 (5) (2022) 06022003,
925 doi:10.1061/(ASCE)HY.1943-7900.0001978.
- 926 [7] Y. Qian, D.Z. Zhu, L. Liu, W. Shao, S. Edwini-Bonsu, F. Zhou, Nu-
927 merical and experimental study on mitigation of storm geysers in Ed-
928 monton, Alberta, Canada, *J. Hydraul. Eng.* 146 (3) (2020) 04019069,
929 doi:10.1061/(ASCE)HY.1943-7900.0001684.
- 930 [8] G. Zhou, A. Prosperetti, Violent expansion of a rising
931 Taylor bubble, *Phys. Rev. Fluids* 4 (7) (2019) 073903,
932 doi:10.1103/PhysRevFluids.4.073903.
- 933 [9] A.S. León, I.S. Elayeb, Y. Tang, An experimental study on violent
934 geysers in vertical pipes, *J. Hydraul. Res.* 57 (3) (2019) 283–294,
935 doi:10.1080/00221686.2018.1494052.
- 936 [10] S.N. Chan, J. Cong, J.H.W. Lee, 3D numerical modeling of geyser forma-
937 tion by release of entrapped air from horizontal pipe into vertical shaft,
938 *J. Hydraul. Eng.* 144 (3) (2018) 04017071, doi:10.1061/(ASCE)HY.1943-
939 7900.0001416.
- 940 [11] J.G. Vasconcelos, S.J. Wright, P.L. Roe, Improved simulation of
941 flow regime transition in sewers: two-component pressure approach,

- 942 J. Hydraul. Eng. 132 (6) (2006) 553–562, doi:10.1061/(ASCE)0733-
943 9429(2006)132:6(553).
- 944 [12] Z. Xue, L. Zhou, D.Z. Zhu, A decoupled two-fluid cut-cell finite-volume
945 framework with heterogeneous Riemann closures for mixed-flow transi-
946 tions, manuscript in preparation (2026).
- 947 [13] R.I. Issa, M.H.W. Kempf, Simulation of slug flow in horizontal and
948 nearly horizontal pipes with the two-fluid model, Int. J. Multiphase
949 Flow 29 (1) (2003) 69–95.
- 950 [14] Y. Taitel, A.E. Dukler, A model for predicting flow regime transitions in
951 horizontal and near horizontal gas–liquid flow, AIChE J. 22 (1) (1976)
952 47–55.